

- 11. The **electrosurgical pencil is kept in its holster** when not in use.
- 12. ESU works at **300 kHz-3 MHz**; **bipolar needs no dispersive pad** and is safest with pacemakers.
- 13. **Autoclave 121 °C / 15 psi / 15-20 min**; **hot air oven 160 °C / 2 h**.
- 14. The **Bowie-Dick test checks air removal** in a prevacuum steriliser; the **biological indicator is the only proof of sterilizing efficacy**.
- 15. **Ethylene oxide requires aeration**; **hydrogen peroxide plasma cannot be used for cellulose**.
- 16. **SAL = 10⁻⁶**; sterility is now **event-related, not time-related**.
- 17. **HEPA = 99.97 % at 0.3 µm**; the OR is under **positive** pressure with **≥ 15 air changes/hour**.
- 18. **Hand hygiene is the single most effective measure** against healthcare-associated infection.
- 19. A **draped table is sterile only at table-top level**; the wrapper edge (**1 inch**) is unsterile; **moisture = contamination**.
- 20. **X-ray-detectable sponges are never used as dressings**, and **nothing leaves the OR until the final count is correct**.
- 21. **Malignant hyperthermia** — triggers are **volatile agents and succinylcholine**; treat with **dantrolene 2.5 mg/kg**; the earliest sign is a **rising EtCO₂**.
- 22. **Anaphylaxis — adrenaline 0.5 mg IM (1:1000)**; **latex reactions are typically delayed 20-60 min**.
- 23. **CPR: 100-120/min, 5-6 cm deep, 30:2**; **VF and pulseless VT are the only shockable rhythms**.
- 24. The commonest **ignition** source of OR fires is the **ESU**; the commonest **fuel** is **alcohol-based prep**.
- 25. **Aldrete ≥ 9** for PACU discharge; **PADSS ≥ 9** for discharge home; **pain is in PADSS, not Aldrete**.
- 26. The commonest immediate PACU problem is **airway obstruction from the tongue**; the commonest post-op pulmonary complication is **atelectasis**.
- 27. **Post-op fever: Wind → Water → Walking → Wound → Wonder drugs**. Fever in the first 24 h is usually **not** infective.
- 28. **Dehiscence POD 5-10**, heralded by **serosanguineous discharge**; **evisceration → cover with warm sterile saline-soaked gauze and call the surgeon**.
- 29. The **colon is the last part of the gut to recover** (48-72 h); return of function is judged by **passage of flatus**.
- 30. **Radiation: 20 mSv/year occupational**; **doubling the distance quarters the exposure**; the **dosimeter is worn OUTSIDE the apron**.

End of supplementary notes • Revise the PDF for Chapters 1-3 core text, and this document for every gap listed in the coverage audit on page 1.

SUPPLEMENTARY NOTES — WHAT YOUR PDF DID NOT COVER

Preparation of Patient for Operation • Pre- & Post-Operative Patient Care • Protection of the Patient in Surgery • OT Safety for the Medical Assistant / Pharmacist

Gap-analysis of the studied PDF (Ch. 1 Perioperative Care of the Patient • Ch. 2 Protection of the Patient in Surgery / Patient Safety • Ch. 3 Safety Measures for Operating Room Personnel) + complete MCQ-proofing

HOW TO USE THIS DOCUMENT

Your PDF is **strong** on: psychological support, patient identification/Universal Protocol, the perioperative record forms, surgical positions, counting procedure, the four sterilization modalities, ESU precautions, orientation / in-service education, body mechanics, fatigue and waste anaesthetic gases. **Do not re-read those in depth — revise them from the PDF.**

This document contains **only the deficits**: every syllabus sub-point where the PDF is silent or gives qualitative text where examiners ask for **numbers, classifications, doses and mechanisms**.

► **Biggest single gap: POST-OPERATIVE PATIENT CARE.** The syllabus heading says “Pre & Post Operative Patient Care”, but the PDF stops at handing the patient over to PACU. Part C is therefore essential. Sources followed: AORN Guidelines for Perioperative Practice; Berry & Kohn's Operating Room Technique; CDC/HICPAC Guidelines (SSI, disinfection & sterilization, isolation precautions); WHO Surgical Safety Checklist & Hand Hygiene guidelines; AAGBI/ASA fasting & MH guidance; ICRP/AERB radiation limits; OSHA/NIOSH exposure limits; Biomedical Waste Management Rules 2016 (India); Bailey & Love; Lawrence's Essential Surgical Specialties.

0. COVERAGE AUDIT — SYLLABUS VS. YOUR PDF

Read this table first. ✓ = safe to revise from the PDF alone. △ = PDF is partial, the exam-relevant detail is in this document. ✗ = essentially absent from the PDF.

Syllabus sub-topic	Status	Where the PDF covers it	What is still missing → read here
Psychological support of the surgical patient	✓ + △	Ch.1, pp. 1-11 (very detailed)	Anxiety assessment scales, coping mechanisms, pre-op teaching models, special groups → §A1
Preparation of patient for operation (physical prep)	✗	Only the checklist form (Form 1-1)	Fasting rules, hair removal, bathing, bowel/bladder prep, premedication, ASA grading, investigations, VTE & antibiotic prophylaxis, ERAS → §A2-A9
Informed consent	△	Forms 1-2, 1-3 reproduced	Legal elements, types, who may consent, age, IPC sections, exceptions, medico-legal doctrines → §A10
Admission procedure / patient identification	✓	Ch.2, pp. 18-19	Only add the WHO Surgical Safety Checklist (3 phases) which the PDF omits → §B1
Transfer procedure & position	✓ + △	Ch.2, pp. 19-30, Fig. 2-1 A-J	Position-specific complications , nerve-injury map, physiological effects, angles/degrees, transfer-device load limits → §B6-B7
Environmental controls	△	Ch.2, pp. 33-35 (qualitative)	OR temperature/humidity/air-change/pressure numbers, HEPA & laminar flow, zoning, cleaning schedule, disinfectant table, SSI classification, hand-hygiene timings, sterile-field rules → §B2-B5

Syllabus sub-topic	Status	Where the PDF covers it	What is still missing → read here
Electro surgery	⚠	Ch.2, pp. 38–39 (5 precautions)	Frequency/waveforms, monopolar vs bipolar circuits, alternate-site burns, capacitive & direct coupling, insulation failure, REM/AEM, pacemaker-ICD protocol, advanced energy devices, surgical smoke → §B8
Operative records	✓ + ⚠	Forms on pp. 3–4, 12–17	Specimen handling & labelling, implant log, incident report rules, retention, what makes a record legally defensible → §B11
Counting procedure	✓	Ch.2, pp. 39–41 (9 rules)	Retained surgical items epidemiology, gossypiboma, RF-detection technology → §B10
Sterilization	⚠	Ch.2, pp. 41–45 (4 modalities)	Definitions & Spaulding classification, exact time-temperature-pressure tables , dry heat, radiation, filtration, physical/chemical/biological indicators, SAL, packaging & shelf-life, levels of disinfection → §B12
Emergencies and disasters	⚠	Ch.2, pp. 47–49 (CPR cart, power failure, MH, death in OR)	BLS/ACLS numbers, anaphylaxis doses, full MH protocol with dantrolene dosing, LAST, airway fire drill, extinguisher classes, RACE/PASS, mass-casualty triage, code colours → §B13
POST-OPERATIVE PATIENT CARE	✗	PACU hand-over paragraph only (p. 8) + PACU form	PACU monitoring, Aldrete & PADSS scores , immediate complications, PONV, pain, post-op fever timeline, wound healing & dehiscence, drains, fluids, DVT, ileus, discharge counselling → entire Part C
In-service education	✓	Ch.3, pp. 50–51	Nothing significant — revise from PDF
Body mechanics	✓ + ⚠	Ch.3, p. 51	NIOSH lifting limits, safe-patient-handling numbers → §D1
Fatigue factors	✓	Ch.3, pp. 51–52	Nothing significant — revise from PDF
Radiation safety	⚠	Ch.3, p. 52 (4 measures)	Units (Gy/Sv/rem), dose limits , ALARA, inverse-square law, lead-equivalence, dosimeter types, stochastic vs deterministic effects, pregnancy limits → §D2
Infection control	⚠	Ch.3, pp. 53–54	Standard vs transmission-based precautions, PPE donning/doffing order, needle-stick transmission risks & PEP, immunisation, biomedical waste colour coding → §D3
Chemical hazards	⚠	Ch.3, pp. 54–56	OSHA/NIOSH exposure limit numbers , glutaraldehyde/formaldehyde/EO, methyl methacrylate, cytotoxic drug handling, SDS & GHS, spill management → §D4

EXAM-STRATEGY NOTE
 Roughly **60–65 %** of MCQs from this unit in Indian paper patterns come from just five clusters: **(1)** sterilization time–temperature-indicator numbers, **(2)** post-operative complications & scoring systems, **(3)** electrosurgery hazards, **(4)** patient positioning nerve injuries, **(5)** radiation / chemical / biomedical-waste numbers.
 Four of those five are the PDF's weak areas. Prioritise **§B12, Part C, §B8, §B6, §D2–D4**.

Mnemonic	Stands for
Sign In - Time Out - Sign Out	The 3 phases of the WHO Surgical Safety Checklist
A-R-C-C-O	Aldrete score: A ctivity, R espiration, C irculation, C onsciousness, O xygen saturation — max 10, discharge ≥ 9
Wind - Water - Walking - Wound - Wonder drugs	Post-op fever by day: 1-2 → 3-5 → 4-6 → 5-7 → > 7
Virchow's triad	S tasis + E ndothelial injury + H ypercoagulability → VTE
R - A - C - E	Fire: R escue, A larm, C onfine, E xtinguish/Evacuate
P - A - S - S	Extinguisher: P ull, A im at the base, S queeze, S weep
R - P - M	START triage: R espiration, P erfusion, M ental status
4 H's & 4 T's	Reversible causes of arrest: Hypoxia, Hypovolaemia, Hypo/Hyperkalaemia-metabolic, Hypothermia; Tension pneumothorax, Tamponade, Toxins, Thrombosis
4 D's	Negligence: D uty, D ereliction, D irect causation, D amage
Critical - Semi-critical - Non-critical	Spaulding: sterilize – high-level disinfect – low-level disinfect
Geo = Steam ; Atro = Air & EO	Ge obacillus stearothermophilus → steam & plasma; Bacillus atro phaeus → dry heat (air) & E thylene oxide
‘Yellow burns, Red recycles, White is sharp, Blue is glass’	Biomedical waste colour coding (BMW Rules 2016)
Time - Distance - Shielding	The three cardinal principles of radiation protection (ALARA)
‘If the gut works, use it’	Enteral is always preferred to parenteral nutrition
‘The treatment of pus is drainage’	Surgical site infection management
‘When in doubt, throw it out’	An item of doubtful sterility is considered unsterile
‘Only what is documented is a legal fact’	Perioperative documentation

The 30 statements most likely to appear as a one-mark question

- The **circulator** keeps the legal record of the count and acts as the **patient's advocate**.
- At least two identifiers** are used; the patient **states** their own name rather than being asked “Are you Mary Jones?”
- The site is marked by the **operating surgeon** with an **indelible marker**, with the patient participating.
- The **‘time out’** occurs **after induction and before skin incision**, and every team member must participate.
- Hearing is the last sense to be lost** — silence is mandatory during induction.
- The **safety strap across the mid-thighs** is the single most important measure against falls from the table.
- Ulnar nerve** injury is the commonest peri-operative nerve injury; **common peroneal** in lithotomy.
- Trendelenburg raises ICP and IOP and lowers FRC**; do not maintain it longer than necessary.
- Never move an anaesthetised patient without the anaesthetist's permission**.
- The **patient's skin must never contact metal** when the ESU is used (alternate-site burn).

Q83. If the distance from a radiation source is doubled, the exposure received is reduced to
(a) One half (b) One quarter (c) One eighth (d) One sixteenth
Ans: (b) One quarter — Inverse square law: intensity $\propto 1/d^2$, so doubling the distance gives one quarter the exposure and tripling it gives one ninth. Practically, stand at least 2 m (6 ft) from the source, or leave the room. Time, distance and shielding are the three cardinal principles, governed by ALARA.

Q84. A personal radiation dosimeter should be worn
(a) Inside the lead apron at waist level (b) Outside the lead apron at chest/collar level (c) On the wrist of the dominant hand (d) Attached to the X-ray machine
Ans: (b) Outside the lead apron at chest/collar level — The primary dosimeter is worn **outside the apron at chest or collar level** so that it records the dose to the unshielded neck, head and lens. A **pregnant worker wears a second badge under the apron at waist level**. **TLD** badges are the commonest type.

Q85. Carcinogenesis following radiation exposure is an example of a
(a) Deterministic effect with a threshold (b) Stochastic effect with no threshold (c) Acute deterministic effect (d) Non-radiation effect
Ans: (b) Stochastic effect with no threshold — **Stochastic** effects (**cancer and heritable mutation**) have **no threshold** — the **probability** rises with dose, not the severity. **Deterministic** effects (erythema, epilation, **cataract**, sterility, marrow suppression) have a **threshold**, and their severity increases with dose.

Q86. The OSHA permissible exposure limit (8-hour TWA) for ethylene oxide is
(a) 0.05 ppm (b) 1 ppm (c) 25 ppm (d) 100 ppm
Ans: (b) 1 ppm — EO = 1 ppm (8-h TWA) with a 15-min excursion limit of 5 ppm. Other limits: **glutaraldehyde ceiling 0.05 ppm, formaldehyde 0.75 ppm, nitrous oxide 25 ppm (NIOSH), halogenated volatiles 2 ppm, methyl methacrylate 100 ppm.**

Q87. Under the Biomedical Waste Management Rules 2016, a used syringe with a fixed needle should be discarded into which colour container?
(a) Yellow bag (b) Red bag (c) White translucent puncture-proof container (d) Blue container
Ans: (c) White translucent puncture-proof container — Sharps, including needles and syringes with fixed needles, go into the WHITE translucent, puncture-proof, leak-proof container, then are autoclaved and shredded/mutilated before encapsulation or foundry disposal. **Red** = contaminated recyclable plastics (tubing, syringes **without** needles); **Yellow** = anatomical, soiled, expired drugs and cytotoxics; **Blue** = glassware and metallic implants.

Q88. Cytotoxic drug waste must be incinerated at a temperature of at least
(a) 800 °C (b) 1000 °C (c) 1200 °C (d) 600 °C
Ans: (c) 1200 °C — $\geq 1200\text{ }^{\circ}\text{C}$, or the waste is returned to the manufacturer. Cytotoxic waste is collected in a **separate yellow container labelled ‘cytotoxic’** — including glass vials contaminated with cytotoxics, which therefore do **not** go into the blue stream.

PART G — RAPID REVISION: MNEMONICS AND LAST-MINUTE SHEET

Mnemonics

Mnemonic	Stands for
2 - 4 - 6 - 8	Pre-op fasting: clear fluids 2 h, breast milk 4 h, formula/light meal 6 h, fatty meal 8 h
7 A's	Premedication aims: Anxiolysis, Amnesia, Analgesia, Antisialagogue, Anti-emetic, Anti-acid, Autonomic attenuation (+ Antibiotic prophylaxis)
4 G's + E	Herbals to stop 2 weeks pre-op: Garlic, Ginger, Ginkgo, Ginseng, Ephedra (+ St John's wort, vitamin E, fish oil)

PART A — PREPARATION OF THE PATIENT FOR OPERATION (PRE-OPERATIVE)

A1. Psychological Support — the frameworks your PDF omits

The PDF describes *what* staff do (telephone call, warm blanket, holding the hand, acting as advocate). Examiners also ask *why* and by *what instrument*.

Concept	Key content
Sources of pre-operative anxiety (classic 7)	Fear of the unknown • fear of death • fear of anaesthesia / not waking up • fear of pain • fear of disfigurement & altered body image • fear of loss of control / dependence • separation from family + financial concern
Assessment tools	STAI (State-Trait Anxiety Inventory) • APAIS (Amsterdam Preoperative Anxiety & Information Scale) • HADS • Visual Analogue Anxiety Scale • for children the m-YPAS (modified Yale Preoperative Anxiety Scale)
Peak anxiety times	On being told surgery is needed • night before • immediately before transfer to OR (highest) • in the holding area
Effects of unrelieved anxiety	↑ catecholamines → tachycardia, hypertension, arrhythmia • ↑ anaesthetic requirement • ↑ post-op pain scores & analgesic need • ↑ PONV • delayed wound healing (cortisol) • longer hospital stay
Evidence-based interventions	Structured pre-operative teaching/counselling (single most effective) • information giving matched to coping style (“monitors” want detail, “blunters” want less) • relaxation & guided imagery • music therapy • parental presence at induction / midazolam premedication in children • pre-op visit to the OR for children • allowing spectacles, hearing aids and dentures to remain until induction
Special groups	Paediatric — fear of separation (1–3 y), fear of mutilation (3–6 y), fear of loss of control (school age); Geriatric — fear of dependency & death, sensory deficits, needs slower, repeated explanation; Cancer/amputation — anticipatory grief, body-image work; Jehovah's Witness — document blood refusal (PDF Form 1-1 “refuses transfusions” box)
Legal-ethical anchor	The patient's autonomy is protected by informed consent; the perioperative nurse/practitioner is the patient's advocate — the only team member whose sole duty is the patient's interest while the patient is anaesthetised

A2. Pre-Operative Fasting (“NPO”) — the PDF only has a blank “NPO since __” line
ASA / ESA / AAGBI minimum fasting intervals before elective anaesthesia (healthy patients, no risk of delayed gastric emptying). Remember as 2-4-6-8.

Ingested material	Minimum fast	Notes
Clear liquids (water, pulp-free juice, black tea/coffee, clear carbohydrate drink)	2 hours	Volume is less important than type; ERAS actively encourages a CHO drink 2 h pre-op
Breast milk	4 hours	Neonates & infants
Infant formula, non-human milk, light meal (toast + clear fluid)	6 hours	Milk behaves like a solid (curdles)
Fried / fatty food, meat, full meal	8 hours	May need longer
Chewing gum / boiled sweet	Stop before OR (do not cancel)	Not an indication to postpone in adults

WHY NPO? — CLINICAL CORRELATION

Purpose = prevent **pulmonary aspiration of gastric contents (Mendelson's syndrome)** during induction, when protective airway reflexes are abolished.
Risk becomes significant when gastric volume is **> 0.4 mL/kg (≈ 25 mL)** and **pH < 2.5**.
Full-stomach / high-risk patients (emergency, trauma, pregnancy ≥ 2nd trimester, obesity, diabetic gastroparesis, bowel obstruction, GERD, opioids) → treat as **not fasted** → rapid sequence induction with cricoid pressure (**Sellick's manoeuvre**), aspiration prophylaxis (H₂ blocker / PPI + sodium citrate ± metoclopramide). Prolonged (“midnight”) fasting is now considered **harmful** — dehydration, hypoglycaemia, insulin resistance, irritability, and it does **not** reduce gastric volume.

A3. Routine Pre-Operative Investigations

The PDF lists “history & physical, haematology, blood chemistry, urinalysis, chest X-ray, EKG”. Modern practice is **selective, not routine** — a classic MCQ discriminator.

Investigation	Indication / current recommendation
Hb / PCV	All major surgery; women of child-bearing age; anticipated blood loss > 500 mL
Blood grouping & cross-match / “group & save”	Per the surgical blood-order schedule; availability confirmed by the circulator before surgery starts
Coagulation profile (PT/INR, aPTT)	Only if bleeding history, liver disease, anticoagulants, or major/cardiac surgery — not routine
Blood sugar / HbA1c	Diabetics, > 40 y, obesity; HbA1c ideally < 8 % for elective surgery
Renal function, electrolytes	> 60 y, hypertension, diabetes, diuretics, renal disease, major surgery
Urinalysis	Urological surgery, implant surgery, symptoms; not routine otherwise
ECG	Age > 40–50 y (institution-specific), cardiac disease, hypertension, diabetes, electrolyte disturbance
Chest X-ray	Cardio-respiratory disease, > 60 y, thoracic surgery, suspected TB/malignancy — not routine in healthy young adults
Pulmonary function tests / ABG	Thoracic resection, severe COPD, obesity-hypoventilation
Pregnancy test	All women of child-bearing potential must be asked; test if any doubt (also a radiation-safety requirement — see §D2)
Viral markers (HBsAg, anti-HCV, HIV)	As per institutional policy; result does not change universal application of Standard Precautions

A4. Risk Stratification — ASA grade, Mallampati, and the scores examiners love

ASA Physical Status Classification (ASA-PS)

Class	Definition	Example
ASA I	Normal healthy patient	Fit non-smoker, no or minimal alcohol
ASA II	Mild systemic disease without functional limitation	Controlled hypertension or diabetes, current smoker, pregnancy, obesity BMI 30–40
ASA III	Severe systemic disease with functional limitation	Poorly controlled DM/HTN, COPD, BMI ≥ 40, MI/CVA > 3 months ago, dialysis
ASA IV	Severe disease that is a constant threat to life	MI/CVA/stent < 3 months , ongoing cardiac ischaemia, sepsis, ejection fraction < 25 %

Q75. Post-operative return of bowel function occurs in which order?

(a) Colon → stomach → small bowel (b) Small bowel → stomach → colon (c) Stomach → colon → small bowel (d) All simultaneously at 72 hours
Ans: (b) Small bowel → stomach → colon — Small intestine within ≈ 24 h → stomach 24–48 h → COLON 48–72 h (last to recover). The clinical marker of resolution is the **passage of flatus or stool** — bowel sounds are unreliable.

Q76. According to the Apfel score, which of the following is NOT a risk factor for PONV?

(a) Female sex (b) Non-smoking status (c) Smoking (d) Expected post-operative opioid use
Ans: (c) Smoking — The four Apfel factors are **female sex, non-smoker, previous PONV or motion sickness, and expected post-operative opioid use** — giving risks of roughly **10/20/40/60/80 %** for 0–4 factors. **Smoking is actually protective.**

Q77. The drug of choice given at the END of surgery for PONV prophylaxis is

(a) Metoclopramide (b) Ondansetron (c) Dexamethasone (d) Atropine
Ans: (b) Ondansetron — Ondansetron 4–8 mg IV, a 5-HT₃ antagonist, is most effective given **at the end of surgery** (watch for **QT prolongation** and headache). **Dexamethasone 4–8 mg is given at induction.** High-risk patients need **agents from 2–4 different classes plus propofol TIVA and opioid sparing.**

Q78. The minimum acceptable post-operative urine output in an adult is

(a) 0.1 mL/kg/h (b) 0.5 mL/kg/h (c) 2 mL/kg/h (d) 5 mL/kg/h
Ans: (b) 0.5 mL/kg/h — > 0.5 mL/kg/h (roughly 30 mL/h in an adult). Post-operative oliguria is **most often hypovolaemia** — assess perfusion and give a fluid challenge — but beware of causing **dilutional hyponatraemia** with hypotonic fluid in the ADH-driven post-operative state.

Q79. Post-operative shivering is most effectively treated with warming plus

(a) Diazepam (b) Pethidine (meperidine) (c) Furosemide (d) Naloxone
Ans: (b) Pethidine (meperidine) — Pethidine 12.5–25 mg IV is the most effective drug (via κ-opioid and α₂ effects); clonidine, dexmedetomidine, tramadol and ondansetron are alternatives. **Always give oxygen**, because shivering can increase oxygen consumption by up to 400 %.

Q80. Secondary haemorrhage after surgery classically occurs on which post-operative day and is due to

(a) Day 0; a slipped ligature (b) Within 24 h; a dislodged clot as blood pressure rises (c) POD 7–14; infection eroding a vessel wall (d) POD 30; a false aneurysm
Ans: (c) POD 7–14; infection eroding a vessel wall — Primary haemorrhage = during surgery; **reactionary** = within 24 h (slipped ligature or a clot dislodged as BP rises and vasoconstriction wears off); **secondary = POD 7–14 from infection eroding the vessel wall.**

Section 9 — Radiation, Chemical Hazards and Biomedical Waste

Q81. The SI unit of equivalent dose of radiation is the

(a) Gray (b) Sievert (c) Becquerel (d) Roentgen
Ans: (b) Sievert — Sievert (Sv) is the unit of **equivalent/effective dose** and is used for all dose limits and personnel monitoring (**1 Sv = 100 rem**). The **Gray** is **absorbed dose** (1 Gy = 100 rad) and the **Becquerel** is **activity**.

Q82. The recommended annual occupational whole-body radiation dose limit is

(a) 1 mSv (b) 20 mSv averaged over 5 years (c) 50 mSv every year without restriction (d) 150 mSv
Ans: (b) 20 mSv averaged over 5 years — 20 mSv per year averaged over 5 consecutive years (100 mSv in 5 years), with a maximum of 50 mSv in any single year. Public limit **1 mSv/year**; lens of the eye **20 mSv/year**; extremities **500 mSv/year**; and after a pregnancy is declared, **≤ 1 mSv to the fetus** (0.5 mSv per month).

Q67. Fever on the second post-operative day is most likely due to

(a) Wound infection (b) Atelectasis (c) Deep vein thrombosis (d) Urinary tract infection
Ans: (b) Atelectasis — Follow the **W's in time order: Wind (atelectasis, POD 1-2) → Water (UTI, POD 3-5) → Walking (DVT, POD 4-6) → Wound (SSI, POD 5-7) → Wonder drugs (> POD 7)**. Fever in the first 24 h is usually the inflammatory response, a drug/transfusion reaction, or malignant hyperthermia.

Q68. Fever appearing on the sixth post-operative day with wound erythema and purulent discharge indicates

(a) Atelectasis (b) Drug fever (c) Surgical site infection (d) Malignant hyperthermia
Ans: (c) Surgical site infection — **SSI classically declares itself on POD 5-7**. Management is to **open and drain the wound, send a culture, debride, pack for healing by secondary intention** and give culture-guided antibiotics — ‘the treatment of pus is drainage’.

Q69. Wound dehiscence most commonly occurs on which post-operative day, and its herald sign is

(a) POD 1-2; bright red bleeding (b) POD 5-10; serosanguineous ‘salmon-pink’ discharge (c) POD 14-21; purulent discharge (d) POD 30; a palpable lump
Ans: (b) POD 5-10; serosanguineous ‘salmon-pink’ discharge — Dehiscence peaks around **POD 5-10 (classically day 7-8)**, when the wound has only 5-10 % of its final tensile strength; the warning sign is a **sudden serosanguineous (salmon-pink) discharge** with a palpable gap.

Q70. The correct immediate nursing action for post-operative wound evisceration is to

(a) Push the viscera back and apply a dry dressing (b) Cover the viscera with sterile gauze moistened with warm sterile normal saline, place the patient in low Fowler's with knees flexed, keep NBM and call the surgeon (c) Sit the patient upright and give oral fluids (d) Apply a tight abdominal binder and discharge
Ans: (b) Cover the viscera with sterile gauze moistened with warm sterile normal saline, place the patient in low Fowler's with knees flexed, keep NBM and call the surgeon — **Never attempt to replace the viscera and never use dry gauze**. Cover with **warm sterile saline-moistened gauze**, position supine/low Fowler's with knees flexed to reduce tension, keep the patient **nil by mouth**, secure IV access, monitor for shock, give analgesia and **prepare for immediate theatre**.

Q71. Sutures on the face are typically removed on day

(a) 3-5 (b) 7-10 (c) 10-14 (d) 14-21
Ans: (a) 3-5 — **Face/neck 3-5 days** (excellent vascularity, cosmetic priority); scalp and trunk **7-10 days**; limbs **10-14 days**; over joints, back, palms and soles **14 days**.

Q72. The tensile strength of a surgical wound at one week is approximately what percentage of normal skin?

(a) 5-10 % (b) 30-50 % (c) 70-80 % (d) 100 %
Ans: (a) 5-10 % — Only **5-10 % at 1 week**, rising to **30-50 % at 1 month**, and reaching a maximum of only **70-80 %** of unwounded skin — the wound never regains full strength.

Q73. Which cell is the key orchestrating cell of the inflammatory phase of wound healing?

(a) Neutrophil (b) Macrophage (c) Fibroblast (d) Platelet
Ans: (b) Macrophage — **Neutrophils** arrive first (24-48 h), but the **macrophage (48-72 h) is the pivotal cell** — it debrides and releases the growth factors that recruit fibroblasts. **Fibroblasts** dominate the proliferative phase (day 4-21), depositing **collagen III**, which is later replaced by **collagen I** during remodelling.

Q74. Which of the following is a CLOSED active suction drain?

(a) Penrose drain (b) Corrugated drain (c) Jackson-Pratt drain (d) Sump drain with an open vent
Ans: (c) Jackson-Pratt drain — The **Jackson-Pratt (bulb)** and **Hemovac/Redivac (spring/high vacuum)** drains are **closed active** systems with a lower infection risk. **Penrose and corrugated drains are passive and open**. A fully re-expanded JP bulb means suction has been lost — re-compress it after emptying.

Class	Definition	Example
ASA V	Moribund, not expected to survive without the operation	Ruptured aneurysm, massive trauma, intracranial bleed with mass effect
ASA VI	Declared brain-dead, organs being removed for donation	Organ harvest
suffix E	Emergency operation	ASA II E = otherwise healthy patient for emergency appendicectomy

HIGH-YIELD TRAPS

ASA grade reflects **only the patient's physiological status** — not the magnitude of surgery, not age, not the anaesthetic risk per se.
Pregnancy = **ASA II** (it is a physiological, not pathological, state but not ‘normal healthy’).
Mallampati grades the *oropharyngeal view* (airway), not overall fitness: **I** soft palate + uvula + fauces + pillars; **II** soft palate + uvula + fauces; **III** soft palate + **base of uvula only**; **IV** soft palate **not** visible. Grades III-IV predict difficult intubation.
Other scores worth a line each: **Goldman / Lee's Revised Cardiac Risk Index** (cardiac risk), **Caprini score** (VTE risk), **Apfel score** (PONV – see §C4), **Aldrete score** (PACU discharge – §C1), **Glasgow Coma Scale** (neuro).

A5. Skin Preparation & Hair Removal — evidence the PDF predates

Question	Current evidence-based answer
Should hair be removed?	No — do not remove hair unless it physically interferes with the procedure or wound closure
If it must be removed, how?	Electric clippers with a single-use head (or depilatory cream). Razors are contraindicated — shaving causes micro-abrasions that get colonised
When?	Immediately before surgery, outside the operating room . Risk of SSI: no removal ≈ clipping < shaving immediately before < shaving the night before (highest)
Pre-operative bath	Whole-body wash with soap or 4 % chlorhexidine gluconate the night before and on the day of surgery
Best intra-operative skin antiseptic	Alcohol-based 2 % chlorhexidine gluconate (e.g. Chloraprep) is superior to aqueous povidone-iodine for clean-contaminated surgery. Povidone-iodine 10 % (1 % free iodine) is the alternative
Alcohol + electrosurgery	Alcoholic preps must be allowed to dry completely (≥ 3 min, longer in hair) and must never pool — pooled alcohol is the classic fuel in OR fires (§B13)
Direction of painting	From the incision line outward in concentric circles/strokes; the sponge is never returned to the clean centre; the most contaminated area (stoma, sinus, axilla, perineum, umbilicus) is prepped last
Contraindications	Chlorhexidine: avoid in middle ear (ototoxic), eye/cornea (keratitis), meninges/neural tissue; iodophors: neonates (thyroid, burns), iodine allergy
Extent of prep	Wide enough to allow extension of the incision, new incisions and drain sites without re-prepping

A6. Surgical Antibiotic Prophylaxis

Parameter	Standard
Timing	Single IV dose within 60 minutes before skin incision (within 120 min for vancomycin and fluoroquinolones, which need slow infusion). For tourniquet surgery, complete infusion before tourniquet inflation
Usual drug	1st/2nd-generation cephalosporin — cefazolin 2 g (3 g if ≥ 120 kg); add metronidazole for colorectal/appendicular; vancomycin/clindamycin if MRSA-colonised or β -lactam allergic
Re-dosing intra-op	If procedure exceeds 2 half-lives of the drug (≈ 4 h for cefazolin) or blood loss > 1500 mL
Duration	Stop within 24 h of surgery end (≤ 48 h for cardiac surgery). Prolonging prophylaxis does not reduce SSI and breeds resistance + C. difficile
Prophylaxis vs therapy	Prophylaxis = clean/clean-contaminated cases and implant surgery. Dirty/infected wounds get therapeutic antibiotics, not prophylaxis
Other proven SSI bundles	Maintain normothermia > 36 °C ; peri-op glucose < 180 mg/dL ; adequate tissue oxygenation (FiO₂ 0.8) ; avoid razors; CHG-alcohol prep

A7. Venous Thromboembolism (VTE) Prophylaxis & Peri-Operative Drug Management

Item	Practice points
Virchow's triad	Stasis + endothelial injury + hypercoagulability — all three are produced by surgery + immobility
Mechanical prophylaxis	Graduated compression (TED) stockings, intermittent pneumatic compression / sequential compression devices , early ambulation. (The PDF's forms mention TED & sequential hose but never explain why.) Contraindicated in peripheral arterial disease
Pharmacological prophylaxis	LMWH (enoxaparin 40 mg SC OD) or UFH 5000 U SC BD/TDS, fondaparinux, or DOACs after major orthopaedic surgery; start 12 h pre- or 6–12 h post-op
Highest-risk surgery	Hip & knee arthroplasty, hip fracture, major pelvic/abdominal cancer surgery, neurosurgery, major trauma, spinal cord injury
Aspirin	Usually continued for secondary cardiovascular prevention; stop 7 days only for very high-bleeding-risk surgery
Clopidogrel / prasugrel / ticagrelor	Stop 5-7 days pre-op (ticagrelor 3–5 d) if bleeding risk outweighs stent thrombosis risk — discuss with cardiology
Warfarin	Stop 5 days ; check INR < 1.5; bridge with LMWH if high thrombotic risk (mechanical valve, recent VTE, AF with prior stroke)
DOACs (dabigatran, rivaroxaban, apixaban)	Stop 24-48 h (longer with renal impairment / high bleed risk)
Metformin	Omit on the morning of surgery (lactic acidosis / contrast risk); restart when eating
Insulin	Reduce/omit short-acting; give ~half to two-thirds of usual basal; start glucose-insulin-potassium infusion for major surgery; monitor capillary glucose hourly
Continue on the day of surgery	β -blockers (abrupt stop $\rightarrow$ rebound ischaemia), statins, anti-epileptics, anti-parkinsonian drugs, inhalers, thyroxine, anti-hypertensives except ACE-I/ARB
Withhold	ACE inhibitors / ARBs on the morning of surgery (refractory hypotension), oral hypoglycaemics, diuretics, herbal supplements 2 weeks before (see box)

Q59. The components of the fire triangle are

(a) Fuel, oxidiser and an ignition source (b) Fuel, water and heat (c) Oxygen, nitrogen and heat (d) Fuel, carbon dioxide and a spark

Ans: (a) Fuel, oxidiser and an ignition source — Fuel + oxidiser + ignition (heat). In the OR the commonest **ignition** source is the **electrosurgical unit (≈ 70 -90 % of OR fires)**, the commonest **fuel** is **alcohol-based skin prep** (or drapes), and the **oxidiser** is oxygen or nitrous oxide.

Q60. For an electrical fire in the operating room, the extinguisher of choice is

(a) Water (b) Foam (c) Carbon dioxide (Class C) (d) Class D dry powder

Ans: (c) Carbon dioxide (Class C) — CO₂ (or ABC dry chemical) for Class C electrical fires — never water. CO₂ is preferred around equipment because it leaves no residue. Classes: **A** ordinary combustibles, **B** flammable liquids, **C** electrical, **D** combustible metals, **K** kitchen fats.

Q61. The acronym RACE, used in the event of a fire, stands for

(a) Rescue, Alarm, Confine, Extinguish/Evacuate (b) Report, Assess, Contain, Escape (c) Remove, Alert, Cover, Exit (d) Rescue, Assess, Call, Evacuate

Ans: (a) Rescue, Alarm, Confine, Extinguish/Evacuate — Rescue - Alarm - Confine - Extinguish/Evacuate. To use an extinguisher: **PASS — Pull** the pin, **Aim** at the base of the fire, **Squeeze**, **Sweep**.

Q62. In START triage, a casualty who is not breathing even after the airway has been opened is tagged

(a) Red (b) Yellow (c) Green (d) Black

Ans: (d) Black — Black = deceased/expectant. **Red** = immediate (RR > 30, absent radial pulse or capillary refill > 2 s, or not obeying commands); **Yellow** = delayed; **Green** = minor/walking wounded. START assesses **RPM — Respiration, Perfusion, Mental status** in under a minute per casualty.

Section 8 — Post-Operative Care

Q63. Which of the following is NOT a component of the modified Aldrete score?

(a) Respiration (b) Circulation (c) Pain (d) Oxygen saturation

Ans: (c) Pain — The modified Aldrete score assesses **Activity, Respiration, Circulation, Consciousness and Oxygen saturation (A-R-C-C-O)**, each 0–2, maximum 10, **discharge ≥ 9** . **Pain and nausea/vomiting belong to the PADSS**, used for discharge home after day surgery.

Q64. The original Aldrete score used which parameter that the modified score replaced with pulse oximetry?

(a) Temperature (b) Skin colour (c) Urine output (d) Bowel sounds

Ans: (b) Skin colour — The 1970 Aldrete score used **skin colour**; the modified score substituted **SpO₂ measured by pulse oximetry**, which is objective and more sensitive.

Q65. The commonest cause of upper airway obstruction in the immediate post-anaesthesia period is

(a) Laryngospasm (b) The tongue falling back against the posterior pharyngeal wall (c) Bronchospasm (d) Vocal-cord paralysis

Ans: (b) The tongue falling back against the posterior pharyngeal wall — Loss of pharyngeal muscle tone in the sedated supine patient allows the **tongue to fall backwards** — recognised by snoring and **see-saw (paradoxical) chest movement**. Treat with **head tilt-chin lift or jaw thrust, lateral position, suction and an oral/nasal airway**.

Q66. The commonest post-operative pulmonary complication is

(a) Pneumonia (b) Atelectasis (c) Pulmonary embolism (d) Pneumothorax

Ans: (b) Atelectasis — Atelectasis — typically causing low-grade fever on **POD 1-2** with reduced basal breath sounds. Treatment is **deep breathing, incentive spirometry (10 breaths hourly), early ambulation, adequate analgesia and chest physiotherapy** — not antibiotics.

Q52. Gamma radiation sterilization is typically carried out with a cobalt-60 source at a dose of
(a) 2.5 kGy (b) 25 kGy (c) 250 kGy (d) 2500 kGy

Ans: (b) 25 kGy — 25 kGy (2.5 Mrad). It is a **cold, industrial-scale** method for pre-packed single-use disposables (syringes, gloves, catheters, sutures, blades, heart valves, bone grafts) and is **not used within hospitals** because of cost and shielding requirements.

Q53. Membrane filtration used to sterilize a heat-labile solution employs a pore size of
(a) 0.45 µm (b) 0.22 µm (c) 1.2 µm (d) 5 µm

Ans: (b) 0.22 µm — 0.22 µm membranes retain bacteria and are used for heat-labile liquids (sera, antibiotic solutions, vaccines, media). Note that filtration **removes but does not kill**, and **does not retain viruses or mycoplasma** (a 0.1 µm filter is needed for those).

Section 7 — Emergencies and Disasters

Q54. Malignant hyperthermia is triggered by

(a) Propofol and nitrous oxide (b) Volatile inhalational agents and succinylcholine (c) Non-depolarising muscle relaxants (d) All local anaesthetics

Ans: (b) Volatile inhalational agents and succinylcholine — Triggers are **all volatile agents (halothane, isoflurane, sevoflurane, desflurane) and the depolarising relaxant succinylcholine. Safe drugs: propofol, thiopentone, benzodiazepines, opioids, nitrous oxide, non-depolarising relaxants and all local anaesthetics.**

Q55. The initial dose of dantrolene in malignant hyperthermia is

(a) 0.5 mg/kg (b) 1 mg/kg (c) 2.5 mg/kg (d) 10 mg/kg immediately

Ans: (c) 2.5 mg/kg — 2.5 mg/kg IV as a bolus, repeated every 5-10 minutes up to 10 mg/kg (occasionally 30 mg/kg). Simultaneously: stop the triggers, hyperventilate with 100 % O₂ at ≥ 10 L/min, cool actively (**stop at 38 °C**), treat hyperkalaemia and arrhythmias, and maintain urine output > 1–2 mL/kg/h.

Q56. The earliest and most sensitive sign of malignant hyperthermia under general anaesthesia is

(a) Hyperthermia (b) An unexplained rise in end-tidal CO₂ (c) Myoglobinuria (d) Hypotension

Ans: (b) An unexplained rise in end-tidal CO₂ — Rising EtCO₂ with tachypnoea and tachycardia is the earliest sign; **masseter spasm** after succinylcholine is the classic clinical clue. **Hyperthermia is a LATE sign —** waiting for it costs lives. The defect is in the **RYR1 ryanodine receptor gene on chromosome 19.**

Q57. The recommended adult intramuscular dose of adrenaline in anaphylaxis is

(a) 0.1 mg of 1:10 000 (b) 0.5 mg of 1:1000 (0.5 mL) into the antero-lateral thigh (c) 1 mg of 1:1000 subcutaneously (d) 5 mg nebulised

Ans: (b) 0.5 mg of 1:1000 (0.5 mL) into the antero-lateral thigh — 0.5 mg IM (0.5 mL of 1:1000) into the antero-lateral thigh, repeated every 5 minutes as needed; 50 µg IV boluses may be titrated by an experienced practitioner with monitoring. Then **rapid IV crystalloid 20 mL/kg**, chlorpheniramine, hydrocortisone and salbutamol.

Q58. Latex anaphylaxis under anaesthesia characteristically differs from drug anaphylaxis in that it

(a) Never causes hypotension (b) Is typically delayed 20–60 minutes after exposure (c) Occurs only in adults (d) Responds only to steroids

Ans: (b) Is typically delayed 20-60 minutes after exposure — Because absorption is across **mucosa or peritoneum**, latex reactions are typically **delayed 20-60 minutes** rather than immediate. Highest-risk groups: **spina bifida, children with multiple operations, and health-care workers.** Cross-reacting foods: **banana, avocado, chestnut, kiwi.**

PHARMACIST'S HIGH-YIELD BOX — herbal products and surgery (“the 4 G's + E”)
Stop **2 weeks** before elective surgery: **Garlic, Ginger, Ginkgo biloba, Ginseng** (all → bleeding / platelet inhibition); **Ephedra** (→ hypertension, arrhythmia, interacts with anaesthetics); **St John's wort** (CYP3A4 induction → drug failure, serotonin syndrome); **Kava & Valerian** (→ potentiate sedatives, withdrawal); **Vitamin E** and **fish oil** (→ bleeding).
This is exactly why the PDF's Form 1-1 insists that medications listed on the front of the chart **include OTC and herbal products.**

A8. Pre-Anaesthetic Medication (Premedication) — pharmacist-relevant

Purpose (“7 A's”)	Drug class	Examples
Anxiolysis / amnesia	Benzodiazepines	Midazolam 0.02–0.05 mg/kg IV / 0.5 mg/kg oral (children), diazepam, lorazepam
Analgesia	Opioids, NSAIDs, paracetamol, gabapentinoids	Fentanyl, morphine; pre-emptive paracetamol 1 g, gabapentin 300–600 mg
Antisialagogue (dry secretions)	Antimuscarinics	Glycopyrrolate 0.2 mg (does not cross BBB — preferred), atropine, hyoscine
Anti-emetic	5-HT ₃ antagonists, steroids, D ₂ blockers, NK ₁	Ondansetron 4–8 mg, dexamethasone 4–8 mg, metoclopramide, aprepitant
Anti-acid / aspiration prophylaxis	H ₂ blockers, PPIs, non-particulate antacid, prokinetics	Ranitidine 150 mg oral / 50 mg IV, pantoprazole 40 mg, sodium citrate 0.3 M 30 mL, metoclopramide 10 mg
Attenuation of autonomic reflexes	β-blockers, α ₂ -agonists, lignocaine	Esmolol, clonidine, dexmedetomidine
Antibiotic prophylaxis	See §A6	Cefazolin 2 g IV within 60 min of incision
Also: allergy prophylaxis	H ₁ + H ₂ blockers + steroid	For latex-sensitive or contrast-allergic patients (see §D3)

A9. Bowel & Bladder Preparation, and ERAS

- **Mechanical bowel preparation** (polyethylene glycol, sodium phosphate) ± oral antibiotics: now used selectively for **colorectal** surgery (oral antibiotics + MBP reduces SSI; MBP alone does not). Causes dehydration and electrolyte loss — hence the PDF's emphasis on fluid-volume nursing focus.
- **Bladder:** patient voids immediately before transfer; catheterise only if indicated (long procedure, pelvic surgery, need for hourly output monitoring). Remove the catheter **within 24 h** — CAUTI prevention.
- **Skin/site marking** by the operating surgeon with an indelible marker, patient awake and participating (§B1).
- **Remove** — dentures/bridges, contact lenses, jewellery (tape a wedding ring if it cannot be removed), nail polish & gel/artificial nails (interfere with pulse oximetry and harbour organisms), hairpins, prostheses; **document the disposition of every valuable** (exactly what Form 1-1 records).
- **ERAS (Enhanced Recovery After Surgery)** — the modern framework the PDF predates: pre-op counselling, no prolonged fasting + **carbohydrate drink 2 h pre-op**, no routine bowel prep, no routine drains/NG tubes, short-acting anaesthesia, regional/multimodal opioid-sparing analgesia, active warming, goal-directed fluids, **early oral feeding and mobilisation within 24 h.** Result: fewer complications and shorter stay.

A10. Informed Consent — legal & medico-legal detail

The five elements of valid consent

- 1. **Disclosure** — diagnosis, nature and purpose of the proposed procedure, its risks and benefits, **alternatives** (including doing nothing) and their risks, and the consequences of refusal.
- 2. **Capacity/competence** — the patient must understand, retain, weigh and communicate a decision.
- 3. **Voluntariness** — free of coercion, fraud, duress or misrepresentation; obtained **before** sedative premedication (note the PDF's Form 1-2 wording: "I am not under the influence of any preoperative or other sedating medication").
- 4. **Comprehension** — in a language the patient understands; use an interpreter, not a family member, where possible.
- 5. **Documentation & authentication** — signed and dated by patient/legal representative, the surgeon (affirmation of informed consent) and a witness; filed in the chart **before** the patient enters the surgical suite.

Types of consent, and who may give it

Type / situation	Rule
Implied consent	Patient's conduct implies agreement (extending an arm for venepuncture). Sufficient for routine examination only
Expressed — oral or written	Written required for any invasive/operative procedure, anaesthesia, blood transfusion, and for research
Informed consent	The legal standard for surgery/anaesthesia (elements above)
Special/additional consent	Sterilization (patient must state that they understand permanent loss of fertility — PDF Form 1-3), organ donation, HIV testing, photography/video, participation in teaching, implant use, blood products, high-risk procedures
Age	18 years and above in India (Indian Majority Act; IPC § 87-88). For a minor or an incompetent adult, the parent/legal guardian consents (IPC § 89 — child under 12 or person of unsound mind)
Emergency / unconscious patient	Doctrine of necessity / implied consent — IPC § 92 : life-saving treatment may proceed in good faith without consent when the patient cannot consent and no representative is available. Document the emergency
Consent obtained by fear or misconception	Invalid — IPC § 90
Who must obtain it	The operating surgeon (or a delegate competent to perform the procedure) — not the nurse or pharmacist. The perioperative practitioner's duty is to verify that a valid, signed, dated consent exists and matches the procedure, site and side, and to report any discrepancy
Withdrawal	Consent may be withdrawn at any time before or during a procedure while the patient is competent
Refusal of blood	Honour it; document on the checklist (PDF Form 1-1 "refuses transfusions") and plan cell-saver / haemostatics / tranexamic acid

Q44. Sterilization by hot air oven requires

(a) 121 °C for 15 min (b) 134 °C for 3 min (c) 160 °C for 2 hours (d) 180 °C for 2 hours

Ans: (c) 160 °C for 2 hours — Dry heat: **160 °C for 2 h**, 170 °C for 1 h, or 180 °C for 30 min. It is the method of choice for **glassware, oils, powders, greases and sharp cutting instruments** (does not blunt or corrode them), but not for rubber, plastics or linen.

Q45. Which item CANNOT be sterilized in a hydrogen peroxide gas plasma (STERRAD) sterilizer?

(a) A fibre-optic cable (b) A stainless-steel instrument (c) Cotton gauze and paper drapes (d) A microsurgical instrument

Ans: (c) Cotton gauze and paper drapes — Cellulose-containing materials — paper, cotton, linen, gauze, dressings — absorb the sterilant and make the cycle ineffective; liquids, powders and long narrow lumens are also unsuitable. Its advantages are a **low temperature (45-50 °C)**, a **~1 h cycle**, and **no aeration** because the by-products are only water and oxygen.

Q46. Sterilization with 2 % activated glutaraldehyde requires an immersion time of

(a) 10 minutes (b) 30 minutes (c) 3 hours (d) 10 hours

Ans: (d) 10 hours — 10 hours for sterilization; high-level disinfection is achieved in **20-45 minutes** (some texts, including your PDF, quote 10 minutes). Items must be **rinsed thoroughly in sterile distilled water**, and the activated solution must be **tested for potency with a test strip** before each use.

Q47. After use, peracetic acid decomposes into

(a) Formaldehyde and water (b) Acetic acid, water and oxygen (c) Chlorine and water (d) Ethylene glycol

Ans: (b) Acetic acid, water and oxygen — Peracetic acid breaks down into **acetic acid (vinegar), water and oxygen**, making it environmentally safe and allowing disposal down an ordinary drain. It works **even in the presence of organic soil** and has a low surface tension, so it reaches lumens.

Q48. According to the Spaulding classification, a flexible gastrointestinal endoscope is a

(a) Critical item requiring sterilization (b) Semi-critical item requiring high-level disinfection (c) Non-critical item requiring low-level disinfection (d) Non-critical item requiring only cleaning

Ans: (b) Semi-critical item requiring high-level disinfection — Flexible endoscopes contact **intact mucous membranes → semi-critical → high-level disinfection** (sterilization preferred where feasible). **Critical** items enter sterile tissue or the vascular system and must be **sterilized**; **non-critical** items touch only intact skin.

Q49. The internationally accepted Sterility Assurance Level (SAL) is

(a) 10⁻³ (b) 10⁻⁴ (c) 10⁻⁶ (d) 10⁻¹²

Ans: (c) 10⁻⁶ — SAL = 10⁻⁶ — a probability of no more than one viable microorganism in one million sterilized items. This is the quantitative definition of 'sterile'.

Q50. Which of the following organisms is MOST resistant to sterilization and disinfection?

(a) HIV (b) Mycobacterium tuberculosis (c) Bacterial spores (d) Prions

Ans: (d) Prions — Order of decreasing resistance: **prions > bacterial spores > coccidia > mycobacteria > small non-lipid viruses > fungi > vegetative bacteria > lipid-enveloped viruses (HIV, HBV — the easiest to kill)**. Prions require **1 N NaOH plus 134 °C for 18 min**.

Q51. The shelf-life of a sterilized wrapped pack is currently determined by

(a) A fixed 7-day limit (b) A fixed 30-day limit (c) Event-related sterility (d) The type of chemical indicator used

Ans: (c) Event-related sterility — Event-related sterility — the pack stays sterile until an event compromises it (wetting, tearing, puncture, broken seal, dropping, excessive handling). The old time-based "sterile for 30 days" rule is obsolete.

Q37. Which statement about surgical smoke (plume) is correct?
(a) A standard surgical mask provides adequate protection (b) Ablating 1 g of tissue is roughly equivalent to smoking 3–6 unfiltered cigarettes (c) It is sterile and harmless (d) A HEPA filter is required in the evacuator, not ULPA
Ans: (b) Ablating 1 g of tissue is roughly equivalent to smoking 3-6 unfiltered cigarettes — Surgical smoke contains over 150 chemicals plus viable cells and viral DNA (HPV transmission to surgeons is documented). 1 g of ablated tissue ≈ 3-6 unfiltered cigarettes. Control = smoke evacuator with an ULPA filter held within 1-2 inches, plus a high-filtration mask or N95 — a standard mask is not enough.

Q38. For an average adult, the maximum recommended continuous pneumatic tourniquet time and pressure for the LOWER limb are approximately
(a) 30 min at 200 mmHg (b) 60 min at 250 mmHg (c) 90–120 min at 300–350 mmHg (d) 3 hours at 400 mmHg
Ans: (c) 90-120 min at 300-350 mmHg — Lower limb: 300-350 mmHg for ≤ 90-120 min; upper limb: 250-300 mmHg for ≤ 60 min. If more time is needed, **deflate for 10-15 minutes** and re-inflate. Record site, pressure and both inflation and deflation times.

Section 6 — Counting, Sterilization and Disinfection

Q39. The commonest retained surgical item is
(a) A needle (b) A surgical sponge/gauze (c) A retractor (d) A guidewire
Ans: (b) A surgical sponge/gauze — Sponges/gauze account for roughly half to two-thirds of retained items, most often in the abdomen/pelvis. A retained gauze encased in granulation tissue is a **gossypiboma (textiloma)**, and is a classic **res ipsa loquitur** claim. Risk factors: emergency surgery, unexpected change of procedure and high BMI.

Q40. Standard gravity-displacement autoclave conditions are
(a) 100 °C at atmospheric pressure for 30 min (b) 121 °C at 15 psi for 15–20 min (c) 134 °C at 15 psi for 30 min (d) 160 °C for 2 hours
Ans: (b) 121 °C at 15 psi for 15-20 min — 121 °C (250 °F) at 15 psi for 15-20 minutes, or 134 °C at 30 psi for 3-3.5 minutes. 160 °C for 2 h is a **hot air oven (dry heat)** cycle.

Q41. Flash (immediate-use) steam sterilization of an unwrapped metal instrument is carried out at
(a) 121 °C for 20 min (b) 132 °C for 3 min (c) 132 °C for 30 min (d) 160 °C for 1 hour
Ans: (b) 132 °C for 3 min — 132 °C (270 °F) for 3 minutes for unwrapped non-porous metal items; **10 minutes** for porous, lumened or complex items. IUSS is **only for an urgently needed single item** — never for implants, whole sets, or for convenience.

Q42. The Bowie-Dick test is performed to detect
(a) Sterilizer air leaks and inadequate air removal in a prevacuum sterilizer (b) Adequacy of the ethylene oxide concentration (c) Spore killing efficacy (d) Steam superheating only
Ans: (a) Sterilizer air leaks and inadequate air removal in a prevacuum sterilizer — The Bowie-Dick is a Class 2 chemical indicator run daily in an EMPTY prevacuum chamber at 134 °C for 3.5 min, and it tests **air removal / steam penetration.** It does **not** prove sterility — only a **biological indicator** does that.

Q43. The biological indicator used to monitor steam sterilization is
(a) Bacillus atrophaeus (b) Geobacillus stearothermophilus (c) Bacillus pumilus (d) Clostridium sporogenes
Ans: (b) Geobacillus stearothermophilus — Geobacillus stearothermophilus (a thermophile, incubated at 55–60 °C) monitors **steam and hydrogen peroxide plasma.** **Bacillus atrophaeus** monitors **dry heat and ethylene oxide;** **Bacillus pumilus** monitors **gamma radiation.**

MEDICO-LEGAL DOCTRINES — one-liners frequently asked
Negligence (Bolam test) — duty of care, breach of standard of a reasonably competent practitioner, causation, damage. The 4 D's: **Duty, Dereliction, Direct** causation, **Damage.**
Res ipsa loquitur ('the thing speaks for itself') — negligence is presumed; the classic example is a **retained surgical sponge or instrument,** wrong-site surgery, or a diathermy burn.
Vicarious liability / respondeat superior — the hospital is liable for its employees' acts.
Captain-of-the-ship doctrine — historically made the surgeon liable for everything in the OR; largely replaced by independent professional accountability of each team member.
Battery — operating without any consent. **Negligence** — operating with inadequately informed consent.
Only what is documented is a legal fact — the single most quoted sentence from your PDF (p. 8, p. 48). An undocumented event 'did not happen' in court.

PART B — PROTECTION OF THE PATIENT IN SURGERY (GAP-FILL)

B1. The WHO Surgical Safety Checklist — completely absent from your PDF
Your PDF teaches the **Joint Commission Universal Protocol (2004)** and the AORN 2005 addition. The **WHO Surgical Safety Checklist (2008, 'Safe Surgery Saves Lives')** is the internationally examined instrument. It has **19 items in 3 phases,** is led by a nominated **checklist coordinator,** and reduced mortality from 1.5 % to 0.8 % and complications from 11 % to 7 % in the original 8-country study.

Phase	When	Core content
1. SIGN IN	Before induction of anaesthesia (with anaesthetist + nurse)	Patient confirms identity, site, procedure, consent • site marked • anaesthesia safety check completed • pulse oximeter on and working • known allergy? • difficult airway/aspiration risk? • risk of blood loss > 500 mL (7 mL/kg in children) → IV access/fluids/blood planned
2. TIME OUT	After induction, before skin incision (whole team pauses)	All members introduce name and role • surgeon, anaesthetist and nurse verbally confirm patient, site, procedure • anticipated critical events (surgeon: critical steps, duration, blood loss; anaesthetist: patient-specific concerns; nurse: sterility indicator confirmed, equipment issues) • antibiotic prophylaxis given in last 60 min? • essential imaging displayed? • VTE prophylaxis • correct position
3. SIGN OUT	Before the patient leaves the operating room	Nurse verbally confirms with the team: name of the procedure recorded • instrument, sponge and needle counts are correct • specimens labelled (including patient name, read aloud) • any equipment problems to be addressed • surgeon/anaesthetist/nurse state the key concerns for recovery and management of this patient

DISTINGUISH THESE THREE — favourite MCQ
Universal Protocol (JC, July 2004) = pre-procedure **verification + site marking + “time out”** immediately before starting. Aim: prevent wrong-site, wrong-procedure, wrong-person surgery.
AORN (March 2005) added confirmation of the **correct patient position** to the time out.
WHO Checklist (2008) = the 3-phase, 19-item tool (Sign in / Time out / Sign out) — broader than site verification; includes airway, blood loss, antibiotics, imaging, counts and specimen labelling.
Site marking rules: by the **operating surgeon,** with an **indelible/permanent marker,** at or near the incision, **with the awake patient participating, before** the patient enters the OR; mark **both** laterality and level (spine); never mark the non-operative side with an 'X'.

B2. Operating Room Design & Environmental Control — the numbers

Parameter	Standard value	Rationale / exam point
Temperature	20-23 °C (68-75 °F)	Comfort of gowned team vs prevention of patient hypothermia ; raise to 24-26 °C for neonates/burns
Relative humidity	20-60 % (classically 30-60 %)	Too low → static electricity/spark risk; too high → microbial growth & condensation
Air changes per hour	Minimum 15 , recommended 20-25 (≥ 3 of them outdoor air)	Dilutes airborne contaminants and waste anaesthetic gas
Air pressure	Positive with respect to corridors and adjacent areas (≥ 2.5 Pa / 0.01 in H ₂ O)	Air flows out of the OR. (Contrast: TB isolation room = negative pressure)
Airflow pattern	Unidirectional, downward from ceiling diffusers over the table, exhaust at floor level (low wall returns, ≥ 2)	Keeps the clean air column over the sterile field
HEPA filtration	High-Efficiency Particulate Air filter = ≥ 99.97 % of particles ≥ 0.3 µm	The number most often asked. ULPA = 99.999 % at 0.1–0.2 µm (used in smoke evacuators)
Laminar (ultraclean) airflow	300-600 air changes/h , unidirectional at 0.3–0.5 m/s	Used for joint arthroplasty and implant surgery ; reduces air bacterial counts to < 10 CFU/m ³
Doors	Kept closed ; traffic minimised	Every door opening breaks the positive-pressure gradient; each additional person adds bacteria-carrying skin scales
Wall/floor finish	Hard, non-porous, seamless, scrubbable; rounded corners; no false ceiling over the OR	Prevents dust retention
Lighting	General 200–300 lux; operative field 20 000-100 000 lux , shadowless, cool (dichroic/LED)	Heat from lights contributes to fluid evaporation from tissue
Noise	Recommended < 45 dB	Noise impairs communication and team performance; silence mandatory during induction (your PDF, p. 8) and during counts
Ideal OR size	≈ 400–600 sq ft (37–60 m ²); cardiac/orthopaedic larger	Space for equipment without breaching the sterile field

Zoning of the operation theatre complex

Zone	Also called	Areas	Attire
Protective / outer zone	Unrestricted	Reception, waiting, changing rooms, offices, pre-op holding, PACU	Street clothes permitted
Clean zone	Semi-restricted	Peripheral corridors, storage, utility, instrument processing, recovery corridor	Scrub attire + head cover (all hair covered); mask not required
Sterile / aseptic zone	Restricted	Operating room, scrub area, sterile store/set-up room	Scrub attire + head cover + mask ; sterile gown & gloves within the sterile field
Disposal / dirty zone	Outer/dirty corridor	Dirty utility, disposal corridor, sluice	Scrub attire; PPE for handling soiled items

Q30. Electrosurgical units operate in which frequency range?

(a) 50–60 Hz (b) 1–10 kHz (c) 300 kHz – 3 MHz (d) 10–100 MHz

Ans: (c) 300 kHz – 3 MHz — Radiofrequency, 300 kHz-3 MHz. Above about **100 kHz** the current no longer stimulates nerve and muscle (the **Faradic effect** is avoided), which is why RF current cuts and coagulates without electrocuting the patient.

Q31. Which of the following is true of a bipolar electrosurgical unit?

(a) It requires a dispersive return electrode pad (b) Current passes through the whole patient (c) No dispersive pad is required (d) It cannot be used near a pacemaker

Ans: (c) No dispersive pad is required — In bipolar electrosurgery both electrodes are the **two tines of the forceps**, so current passes only through the tissue between them. **No pad is needed**, lateral spread is minimal, and it is the **safest mode in patients with pacemakers**.

Q32. The coagulation waveform of an electrosurgical generator is characterised by

(a) Continuous low-voltage current (b) Interrupted high-voltage current (c) Direct current (d) Continuous high-frequency current at 50 Hz

Ans: (b) Interrupted high-voltage current — Coag = interrupted, high-voltage, low duty cycle (≈ 6 %); cut = continuous, low-voltage, high-current (100 % duty cycle). The high voltage of coag mode is why it carries the greatest risk of **capacitive coupling and insulation breakdown**.

Q33. Capacitive coupling during laparoscopic electrosurgery is MOST likely when

(a) A bipolar forceps is used (b) A metal instrument passes through a plastic (hybrid) trocar cannula (c) The dispersive pad is placed on the thigh (d) The generator is in cut mode at low power

Ans: (b) A metal instrument passes through a plastic (hybrid) trocar cannula — A hybrid metal-plastic trocar system allows current to be induced through intact insulation into surrounding tissue. Use **all-metal or all-plastic** systems, lowest effective power, cut rather than coag mode, short activations, and **active-electrode monitoring (AEM)**.

Q34. A pin-hole break in the insulation of a laparoscopic electrosurgical instrument classically presents as

(a) An immediate visible spark (b) A return-electrode burn (c) Delayed bowel perforation several days later (d) Failure of the generator to activate

Ans: (c) Delayed bowel perforation several days later — Insulation failure discharges current **outside the surgeon's field of view**, causing a full-thickness bowel burn that perforates **3-10 days post-operatively** with peritonitis. Inspect insulation before every use and use AEM instruments.

Q35. A return-electrode contact quality monitoring (REM) system on a modern ESU prevents

(a) Alternate-site burns from metal contact (b) Pad-site burns from inadequate pad contact (c) Capacitive coupling (d) Surgical smoke exposure

Ans: (b) Pad-site burns from inadequate pad contact — A split/dual pad continuously measures impedance between its two halves and **shuts the generator down if contact becomes inadequate**, virtually eliminating **return-electrode (pad-site) burns**.

Q36. The surgeon repeatedly asks the circulator to increase the power on the electrosurgical unit. The correct action is to

(a) Increase the power as requested (b) Change to bipolar mode (c) Stop, check all connections and the dispersive pad, and replace the unit if faulty (d) Reposition the ECG electrodes

Ans: (c) Stop, check all connections and the dispersive pad, and replace the unit if faulty — Repeated requests for more power indicate a **fault — poor pad contact, a loose connection, a frayed cord or a failing generator**. Check the circuit; if no cause is found, **turn off and unplug the unit, label it, obtain another, and document the occurrence and the serial numbers** in the operative record.

Q23. A draped instrument table is considered sterile

(a) Down to 30 cm below the table top (b) Only at and above the level of the table top (c) Down to the floor (d) Only in the central one third

Ans: (b) Only at and above the level of the table top — Anything below the table-top level is unsterile and is never brought back up. Related rules: the wrapper edge (2.5 cm/1 inch) is unsterile, the gown back is unsterile, and moisture causes contamination by strike-through.

Q24. An elective cholecystectomy without spillage is classified as which type of wound?

(a) Clean (b) Clean-contaminated (c) Contaminated (d) Dirty

Ans: (b) Clean-contaminated — Entry into the biliary/GI tract under controlled conditions without unusual contamination = clean-contaminated (Class II), expected SSI 3-10 %. Clean < 2 %, contaminated 15-20 %, dirty/infected 30-40 %.

Q25. A surgical site infection occurring in a patient with an implant is defined by the CDC as an infection within

(a) 7 days (b) 30 days (c) 90 days (d) 1 year

Ans: (c) 90 days — Superficial incisional SSI = within 30 days; deep incisional and organ/space SSI = within 30 days, or 90 days when an implant is in place.

Q26. Which organism is the commonest cause of surgical site infection?

(a) Escherichia coli (b) Pseudomonas aeruginosa (c) Staphylococcus aureus (d) Clostridium perfringens

Ans: (c) Staphylococcus aureus — Staphylococcus aureus (including MRSA) is the commonest, followed by coagulase-negative staphylococci, Enterococcus and E. coli. The commonest source of contamination is the patient's own endogenous flora.

Q27. Alcohol-based hand rub is INEFFECTIVE and soap and water must be used when dealing with

(a) Methicillin-resistant Staphylococcus aureus (b) Clostridioides difficile (c) Influenza virus (d) Pseudomonas aeruginosa

Ans: (b) Clostridioides difficile — Alcohol is not sporicidal, so C. difficile spores require mechanical removal by soap and water (plus contact precautions and hypochlorite for the environment). Hands that are visibly soiled also require soap and water.

Q28. Artificial fingernails are prohibited for operating-room personnel because they

(a) Tear surgical gloves (b) Harbour Gram-negative bacteria and fungi (c) Interfere with pulse oximetry (d) Absorb chlorhexidine

Ans: (b) Harbour Gram-negative bacteria and fungi — AORN prohibits artificial nails and chipped polish because the subungual space harbours Gram-negative bacilli and fungi and cannot be adequately decontaminated. (Nail polish does also interfere with pulse oximetry — but the infection-control reason is the one asked.)

Q29. Which of the following correctly describes the sequence of DOFFING personal protective equipment?

(a) Gown → gloves → mask → goggles (b) Gloves → goggles → gown → mask (c) Mask → gown → gloves → goggles (d) Goggles → gloves → mask → gown

Ans: (b) Gloves → goggles → gown → mask — Doffing: gloves → goggles/face shield → gown → mask/respirator (outside the room) → hand hygiene. The most contaminated item (gloves) comes off first and the respirator last. Donning is the reverse order: gown → mask → goggles → gloves.

Section 5 — Electrosurgery and Equipment

Cleaning schedule (“fumigation” is obsolete — know why)

- **Between cases (‘turnover’ clean):** all horizontal surfaces, the table, arm boards, positioning devices, lights, ESU, suction and the floor **within 1-1.5 m of the field** are wiped with a hospital-grade **germicide/disinfectant**. Your PDF's key teaching: **every case is cleaned identically** — the old practice of special ‘ritual’ cleaning only after ‘dirty’ cases is **obsolete**, because **all** cases are treated as potentially infectious (Standard Precautions).
- **Terminal (daily/end-of-day) clean:** entire room, ceiling, walls, all equipment wheels, cabinets, scrub sinks, corridors, storage; **wet vacuum** the floor.
- **Weekly/monthly:** ducts and filters checked, scrub sinks descaled, cabinets emptied, sterilizer maintenance.
- **Formaldehyde fumigation is no longer recommended** (Group 1 carcinogen, mucosal irritant, requires the room to be sealed and idle for 24-48 h, and does not penetrate crevices reliably). Modern alternatives if no-touch disinfection is needed: **vaporised hydrogen peroxide (VHP), hydrogen peroxide + silver, chlorine dioxide, UV-C irradiation** — always as an **adjunct after** physical cleaning, never instead of it.
- **Bacteriological surveillance:** settle plates / air sampling only for outbreak investigation or after new construction — **routine environmental culturing is not recommended**.
- **Blood spill:** absorb, then disinfect with **1 % sodium hypochlorite** (10 000 ppm available chlorine) for 10 min; large spill = 1 % for 30 min. Wear gloves.

B3. Antiseptics & Disinfectants — levels and agents

Level	What it kills	Agents	Typical use
Sterilization	All microbes including bacterial spores	Steam, EO, H ₂ O ₂ plasma, glutaraldehyde 10 h, peracetic acid, radiation	Critical items (SB12)
High-level disinfection (HLD)	All microbes except large numbers of spores	Glutaraldehyde 2 % (20-45 min), OPA 0.55 % (12 min) , H ₂ O ₂ 7.5 % (30 min), peracetic acid, pasteurisation 75 °C/30 min	Semi-critical — flexible endoscopes, laryngoscope blades, respiratory equipment
Intermediat e-level	Vegetative bacteria, M. tuberculosis , most viruses & fungi; not spores	Alcohols 60-90 %, hypochlorite, iodophors, phenolics	Non-critical surfaces with visible contamination, thermometers
Low-level	Most vegetative bacteria, some fungi/viruses; not TB, not spores	Quaternary ammonium compounds, low-conc. iodophors	Non-critical — floors, walls, BP cuffs, furniture

Antiseptic / disinfectant	Mechanism	Spectrum & key points
Alcohol (ethyl 70 %, isopropyl 60-90 %)	Protein denaturation	Rapid, broadest immediate kill; no residual activity; not sporicidal; inactivated by organic matter; flammable — fire risk; not for open wounds/mucosa
Chlorhexidine gluconate (2-4 %)	Disrupts cytoplasmic membrane	Excellent Gram-positive, good Gram-negative; persistent residual activity , not inactivated by blood; ototoxic and corneal-toxic ; poor sporicidal/mycobacterial
Povidone-iodine 10 % (= 1 % available iodine)	Iodination/oxidation of proteins	Broad, includes spores with prolonged contact; needs 2-3 min contact & must dry to release free iodine; inactivated by blood/pus ; thyroid absorption in neonates; iodine allergy

Antiseptic / disinfectant	Mechanism	Spectrum & key points
Hexachlorophene 3 %	Bacteriostatic – enzyme inhibition	Gram-positive only; cumulative effect; neurotoxic in neonates (avoid)
Sodium hypochlorite	Oxidising, releases free chlorine	Broad including HIV/HBV & spores; corrosive to metals ; inactivated by organic matter; never mix with acids or ammonia (chlorine gas)
Glutaraldehyde 2 %	Alkylation of protein/nucleic acid	Sterilant/HLD; does not corrode lenses or metal → endoscopes; irritant, sensitiser, asthma — needs closed system + ventilation
Hydrogen peroxide	Free-radical oxidation	3 % antiseptic, 7.5 % HLD, 58 % vapourised for plasma sterilization; by-products water + oxygen (eco-friendly)
Ethylene oxide	Alkylation of DNA/RNA/protein	Low-temperature sterilant; toxic, mutagenic, carcinogenic, flammable/explosive ; requires aeration
Formaldehyde	Alkylation	Sterilant/HLD; carcinogen; historically used for fumigation — now discouraged
Quaternary ammonium compounds	Membrane-active cationic surfactant	Low-level only; inactivated by soap, hard water and gauze; environmental surfaces

B4. Surgical Site Infection (SSI) — classification your PDF never gives

Wound classification (Altemeier / CDC) and expected infection rate

Class	Definition	Examples	SSI risk
I - Clean	Uninfected, no inflammation; respiratory, GI, GU and genital tracts not entered ; closed primarily; elective	Thyroidectomy, hernia repair, breast lump, joint replacement	< 2 %
II - Clean-contaminated	Respiratory/GI/GU/genital tract entered under controlled conditions without unusual contamination	Elective cholecystectomy, elective bowel resection, hysterectomy, appendectomy (non-perforated)	≈ 3-10 %
III - Contaminated	Open, fresh, accidental wound; major break in sterile technique ; gross spillage from GI tract; acute non-purulent inflammation	Fresh traumatic wound, enterotomy with spillage, acute appendicitis with inflammation	≈ 15-20 %
IV - Dirty / infected	Old traumatic wound with retained devitalised tissue; existing clinical infection or perforated viscus	Faecal peritonitis, drainage of abscess, perforated appendix	≈ 30-40 %

CDC classification of SSI by depth

- **Superficial incisional** — skin & subcutaneous tissue, within **30 days** of surgery.
- **Deep incisional** — fascia & muscle layers, within **30 days**, or within **90 days if an implant** is in place.
- **Organ/space** — any part opened or manipulated during surgery (e.g. intra-abdominal abscess, mediastinitis, meningitis), within 30/90 days.
- Overall, SSI is the **most common healthcare-associated infection in surgical patients** and the **2nd-3rd most common HAI overall**; commonest organism = **Staphylococcus aureus** (then coagulase-negative staphylococci, Enterococcus, E. coli).

Q15. The Kraske (jackknife) position is a modification of the

- (a) Supine position (b) Prone position (c) Lateral position (d) Fowler's position

Ans: (b) Prone position — Kraske/jackknife is a prone modification with the table flexed at the hips to expose the sacrococcygeal and anorectal area. Trendelenburg, reverse Trendelenburg, Fowler's and lithotomy are all modifications of **supine**.

Q16. When taking a patient out of the lithotomy position, both legs must be lowered slowly and simultaneously chiefly to prevent

- (a) Nerve traction injury (b) A sudden fall in blood pressure (c) Wound dehiscence (d) Deep vein thrombosis

Ans: (b) A sudden fall in blood pressure — Lowering the legs abruptly shifts blood into the lower limbs → **sudden reduction in venous return and a precipitous drop in blood pressure**. Simultaneous, slow lowering by two persons also protects the spine and hips.

Q17. In the lateral position, the axillary (chest) roll is placed

- (a) Directly in the axilla (b) Just caudal to the axilla, under the upper chest wall (c) Under the iliac crest (d) Between the knees

Ans: (b) Just caudal to the axilla, under the upper chest wall — It must be placed **caudal to (below) the axilla** to lift the chest wall and relieve pressure on the **dependent brachial plexus and axillary vessels**. Placing it **in** the axilla causes the very injury it is meant to prevent.

Q18. Post-operative visual loss due to ischaemic optic neuropathy is most associated with which position?

- (a) Supine (b) Lithotomy (c) Prone (d) Sitting

Ans: (c) Prone — The **prone position** (especially long spinal surgery with hypotension, anaemia and venous congestion) carries the highest risk of **POVL**, along with corneal abrasion. The head must be kept neutral with the eyes free of all pressure.

Section 4 — Environment, Asepsis and Surgical Site Infection

Q19. A HEPA filter used in operating-room air conditioning removes

- (a) 99.97 % of particles ≥ 0.3 µm (b) 99.9 % of particles ≥ 1 µm (c) 95 % of particles ≥ 0.5 µm (d) 99.999 % of particles ≥ 0.1 µm

Ans: (a) 99.97 % of particles ≥ 0.3 µm — HEPA = 99.97 % of particles of 0.3 µm or larger. The 99.999 % at 0.1 µm figure describes an **ULPA** filter, used in surgical-smoke evacuators.

Q20. The operating room should be maintained at which pressure relative to the adjacent corridor?

- (a) Negative pressure (b) Positive pressure (c) Equal pressure (d) Alternating pressure

Ans: (b) Positive pressure — Positive pressure ensures air flows **out** of the OR, keeping contaminants out. **Negative** pressure rooms are used for **airborne isolation (tuberculosis, measles, varicella)** with ≥ 12 air changes/hour.

Q21. The recommended relative humidity in an operating room is

- (a) Below 20 % (b) 20–60 % (c) 70–80 % (d) Above 80 %

Ans: (b) 20-60 % — 20-60 % (classically 30–60 %). **Too low favours static electricity and sparks; too high favours microbial growth, condensation and pack strike-through**. Temperature is kept at 20–23 °C with ≥ 15 (ideally 20–25) air changes per hour.

Q22. The single most effective measure for preventing healthcare-associated infection is

- (a) Prophylactic antibiotics (b) Hand hygiene (c) Ultraviolet irradiation of the OR (d) Routine fumigation

Ans: (b) Hand hygiene — Hand hygiene with an antimicrobial agent is the single greatest factor in preventing nosocomial infection (stated explicitly in your PDF, p. 34). Routine formaldehyde fumigation is obsolete, and UV is only a surface/air adjunct.

Q7. Which of the following is confirmed during the “Sign Out” phase of the WHO checklist?

- (a) Pulse oximeter functioning (b) Site marking (c) Correctness of instrument, sponge and needle counts (d) Anticipated blood loss

Ans: (c) Correctness of instrument, sponge and needle counts — Sign Out covers the name of the procedure recorded, correctness of the counts, correct labelling of specimens, equipment problems, and key concerns for recovery. Pulse oximetry and site marking belong to Sign In; anticipated blood loss is discussed at Sign In/Time Out.

Q8. A specimen sent for frozen section should be transported

- (a) In 10 % formalin (b) Fresh, in saline-moistened gauze (c) In absolute alcohol (d) In glutaraldehyde

Ans: (b) Fresh, in saline-moistened gauze — Frozen sections must be sent FRESH — never in formalin, which precludes further processing and immunohistochemistry. Routine histopathology goes into 10 % neutral buffered formalin at about 10× the specimen volume; specimens for culture go into a sterile container with no fixative.

Q9. The legal responsibility for recording the surgical count rests with the

- (a) Surgeon (b) Scrub person (c) Circulator (d) Anaesthesia provider

Ans: (c) Circulator — Both the scrub person and the circulator count aloud together, but keeping the written record of the count is the legal responsibility of the circulator, who also initials any additions.

Section 3 — Positioning and Transfer

Q10. The single most important measure to prevent a patient falling from the operating table is

- (a) Keeping the side rails up (b) A safety (restraint) strap firmly applied across the mid-thighs (c) Padded armboards (d) Two staff at the bedside

Ans: (b) A safety (restraint) strap firmly applied across the mid-thighs — The safety strap across the mid-thighs, about 3 inches (7.5 cm) above the knees, applied firmly but not so tightly as to impair venous return, is the key measure. The patient is also never left unattended in the OR.

Q11. Which nerve is most commonly injured in the lithotomy position, resulting in foot drop?

- (a) Femoral nerve (b) Sciatic nerve (c) Common peroneal nerve (d) Obturator nerve

Ans: (c) Common peroneal nerve — The common peroneal nerve is compressed against the fibular head by the stirrup post → foot drop and loss of sensation over the dorsum of the foot. Adequate padding at the fibular neck prevents it.

Q12. The commonest peripheral nerve injury associated with the supine position is that of the

- (a) Radial nerve (b) Ulnar nerve (c) Median nerve (d) Axillary nerve

Ans: (b) Ulnar nerve — Ulnar neuropathy is the commonest peri-operative nerve injury overall. Prevention: arms abducted ≤ 90°, elbows padded, palms supinated (up), no pressure at the medial epicondyle.

Q13. In the Trendelenburg position, all of the following occur EXCEPT

- (a) Increased intracranial pressure (b) Increased venous return (c) Increased functional residual capacity (d) Cephalad displacement of the diaphragm

Ans: (c) Increased functional residual capacity — Trendelenburg decreases FRC and compliance because the abdominal contents push the diaphragm cephalad → atelectasis. It increases venous return, CVP, ICP and IOP, and may cause facial/laryngeal oedema and tube migration.

Q14. Reverse Trendelenburg position is typically used for

- (a) Abdominal hysterectomy (b) Thyroidectomy (c) Haemorrhoidectomy (d) Nephrectomy

Ans: (b) Thyroidectomy — Reverse Trendelenburg (head up) is used for neck procedures — thyroidectomy, parathyroidectomy, scalene node biopsy — and laparoscopic cholecystectomy. Trendelenburg is used for pelvic surgery such as abdominal hysterectomy; lithotomy/jackknife for anorectal surgery; lateral kidney for nephrectomy.

- Sources of contamination: the patient's own endogenous flora is the commonest source; then the surgical team (shedding, breaks in technique), the environment/air, and instruments.
- Risk factors — patient: age, diabetes, obesity, smoking, malnutrition/hypoalbuminaemia, steroids, immunosuppression, remote infection, colonisation with S. aureus, ASA ≥ 3; operative: duration > 2 h (or > 75th percentile — the T-time), wound class, hypothermia, hypoxia, blood transfusion, foreign body/implant, drains, poor haemostasis, dead space, shaving with a razor.

B5. Aseptic Technique, Scrub, Gowning & the Sterile Field — hard rules

Surgical hand antisepsis

Item	Standard
Agents	4 % chlorhexidine gluconate or 7.5 % povidone-iodine scrub, or an alcohol-based surgical hand rub (which has better immediate kill and is now preferred by WHO)
Duration	First scrub of the day 3-5 minutes (per manufacturer); subsequent scrubs 3 min. Older ‘counted brush-stroke’ method = 30 strokes to nails, 20 per surface
Sequence	Fingertips/nails → fingers & interdigital spaces → hands → wrists → forearms to 5 cm above the elbow; always cleanest (hand) held highest; rinse from fingertips downwards to elbow; never return to a cleaned area
Before scrubbing	Remove all rings, watches, bracelets; nails short, clean, no polish, no gel, no artificial nails (AORN prohibits artificial nails — harbour Gram-negatives & fungi); mask and cap adjusted; eyewear in place; no scrubbing with cuts, abrasions or dermatitis
After scrubbing	Hands held above the elbows and in front of the body, above waist and below shoulder level; dry with a sterile towel (one end per hand, blot from fingers to elbow, never back)
WHO “5 Moments” of hand hygiene	1 before touching a patient, 2 before a clean/aseptic procedure, 3 after body-fluid exposure risk, 4 after touching a patient, 5 after touching patient surroundings
Alcohol rub vs soap & water	Alcohol rub is preferred routinely; soap and water is mandatory when hands are visibly soiled and for Clostridioides difficile and spore-formers (alcohol is not sporicidal)
Double gloving	Recommended for orthopaedic/high-risk cases and known blood-borne infection; perforation indicator (coloured under-glove) shows breaches; change gloves every 90–150 min and after every case

Principles of the sterile field (AORN) — rote-learn these

1. Only sterile items are used within a sterile field.
2. Sterile persons are gowned and gloved; the gown is sterile in front from chest to the level of the sterile field, and the sleeves from 5 cm above the elbow to the cuff. The back is never considered sterile.
3. A draped table is sterile only at table-top level; anything hanging below the table edge is unsterile and is never brought back up.
4. Sterile persons keep hands at or above waist level and in front of the body; they pass back-to-back or face-to-face.
5. Unsterile persons stay at least 30 cm (1 ft) away from the sterile field, never reach over it, and always face it.
6. The edge of any sterile wrapper, package or basin is considered unsterile — conventionally a 2.5 cm (1 inch) margin.
7. Moisture causes contamination by capillary action (strike-through); a wet sterile field is a contaminated field.

8. Sterile packages are opened **farthest corner first, nearest corner last**, holding the pack away from the body (exactly your PDF, pp. 35, 42) — so you never reach over the sterile contents.
9. Sterile items must be kept **in view**; anything of **doubtful sterility is considered unsterile** (‘when in doubt, throw it out’).
10. Movement in and around the field is kept minimal; talking is minimised (droplet dispersal); the number of personnel is kept low.

B6. Positioning — complications, angles and the nerve-injury map

Your PDF describes each position beautifully but gives few complication specifics. **Positioning injuries are the second commonest source of anaesthesia-related claims after airway injury**, and peripheral nerve injury is the most examined item.

Position	Nerve/structure at risk	Mechanism & prevention	Physiological effect
Supine	Ulnar nerve (commonest nerve injury overall); brachial plexus; radial n. (armboard edge); occiput/sacrum/heel pressure	Arms abducted ≤ 90° , palms up (supination) , elbows padded; do not let the arm slip off the armboard; pad occiput, sacrum, heels; legs uncrossed	↓ FRC; aorto-caval compression in late pregnancy (→ left lateral tilt 15°)
Trendelenburg	Brachial plexus (shoulder braces), retinal & cerebral congestion, sliding	Padded shoulder braces over the acromio-clavicular area, never on the soft tissue of the neck; limit to 30–45° and to the shortest time possible	↑ venous return , ↑ ICP , ↑ IOP , ↑ CVP ; ↓ FRC & compliance (cephalad diaphragm) → atelectasis; facial/laryngeal oedema; risk of endotracheal tube migration and regurgitation
Reverse Trendelenburg / sitting (Fowler's)	Sciatic stretch, footdrop, venous pooling; venous air embolism in sitting craniotomy	Padded footboard, antiembolism stockings + sequential compression device , slow position change, secure head in traction device (Crutchfield/Mayfield)	↓ venous return → hypotension ; ↓ cerebral perfusion pressure (≈ 2 mmHg per 2.5 cm of height above heart) ; air embolism risk
Lithotomy	Common peroneal nerve (commonest in lithotomy → foot drop) ; femoral, obturator, saphenous, sciatic; lower-limb compartment syndrome	Pad the fibular head away from the stirrup post; raise and lower both legs together, slowly, by two people ; avoid extreme hip abduction/external rotation and hyperflexion; limit time (risk rises sharply beyond 2-4 h) ; protect fingers when the foot of the table is raised	↑ preload initially; on lowering the legs → sudden ↓ venous return → hypotension (exactly why your PDF says lower both legs slowly and simultaneously); ↓ FRC
Prone	Brachial plexus, ulnar, eye → corneal abrasion and post-operative visual loss (POVL) from ischaemic optic neuropathy or central retinal artery occlusion ; breasts, genitalia, iliac crests; neck	Chest/bolster rolls from clavicle to iliac crest to free the abdomen and allow diaphragm excursion; head on a padded horseshoe/donut with the eyes free of pressure and taped closed; keep head neutral, avoid direct pressure on the eyes; check ET tube after turning	↑ abdominal & epidural venous pressure → bleeding; ↓ cardiac index; airway displacement risk; improves oxygenation (basal recruitment)

ABOUT THESE QUESTIONS

These **88 questions** are modelled on the pattern and difficulty of previous-year questions in Indian competitive and university papers that examine this unit (GPAT/NIPER-style pharmacy papers, nursing and OT-technology papers, and PG entrance patterns). They are **written for this document from the standard sources listed on page 1** — they are **not** verbatim reproductions of any single paper, so **use them as coverage checks, not as a leaked question bank**.
Deliberate weighting, matched to where the marks fall: **post-operative care 18 • sterilization & counting 15 • asepsis/environment/SSI 11 • electrosurgery & equipment 9 • positioning 9 • emergencies 9 • radiation/chemical/waste 8 • identification & records 5 • pre-op & consent 4**.
Every question is answered with the reasoning and the distractor logic, so a wrong attempt still teaches you the three facts around it.

Section 1 — Pre-Operative Preparation and Consent

Q1. As per ASA guidelines, the minimum fasting period for clear liquids before elective surgery in a healthy adult is

- (a) 1 hour (b) 2 hours (c) 4 hours (d) 6 hours

Ans: (b) 2 hours — Remember **2-4-6-8**: clear fluids 2 h, breast milk 4 h, formula/light meal 6 h, fatty meal 8 h. Prolonged fasting is harmful and does not reduce gastric volume.

Q2. The preferred method of hair removal before surgery is

- (a) Shaving with a razor the night before (b) Shaving with a razor in the OR (c) Electric clippers immediately before surgery (d) No removal is ever permitted

Ans: (c) Electric clippers immediately before surgery — Ideally hair is **not removed at all**; if it interferes with the procedure, use **clippers immediately before surgery, outside the OR**. Razor shaving — especially the night before — causes micro-abrasions and carries the **highest SSI risk**.

Q3. Consent for a surgical procedure on a 10-year-old child must be obtained from the guardian under which section of the IPC?

- (a) Section 87 (b) Section 88 (c) Section 89 (d) Section 92

Ans: (c) Section 89 — § 89 covers acts done in good faith for the benefit of a **child under 12 years** or a person of unsound mind, by or with the consent of the guardian. § 87–88 relate to consent by persons above 18; **§ 92** covers acts done in good faith **without consent** in an emergency.

Q4. A patient with well-controlled hypertension and no functional limitation is classified as

- (a) ASA I (b) ASA II (c) ASA III (d) ASA IV

Ans: (b) ASA II — ASA II = mild systemic disease **without** functional limitation. Pregnancy, current smoking and BMI 30–40 are also ASA II. Functional limitation moves the patient to ASA III.

Section 2 — Patient Identification, Safety Checklist and Records

Q5. The WHO Surgical Safety Checklist consists of how many phases?

- (a) Two (b) Three (c) Four (d) Five

Ans: (b) Three — Three: Sign In (before induction), **Time Out** (after induction, before skin incision) and **Sign Out** (before the patient leaves the OR) — 19 items in total.

Q6. The surgical “time out” must be performed

- (a) On admission to the ward (b) In the holding area before transfer (c) After induction of anaesthesia but before skin incision (d) Immediately after wound closure

Ans: (c) After induction of anaesthesia but before skin incision — The time out is the whole-team verbal pause **immediately before incision** (before the scalpel is passed). The **Universal Protocol** was mandated by the Joint Commission in **July 2004**; AORN added confirmation of **correct position** in March 2005.

Topic	Number to remember
Tourniquet	Upper limb 250-300 mmHg, ≤ 60 min • lower limb 300-350 mmHg, ≤ 90-120 min ; overlap 3-6 inches ; deflate 10-15 min if longer
CPR	Rate 100-120/min , depth 5-6 cm , ratio 30:2 , biphasic shock 120-200 J , adrenaline 1 mg q3-5 min , amiodarone 300 mg
Anaphylaxis	Adrenaline 0.5 mg IM (0.5 mL of 1:1000) , repeat q5 min; IV 50 µg boluses; fluid 20 mL/kg
Malignant hyperthermia	Dantrolene 2.5 mg/kg IV, up to 10 (-30) mg/kg ; O ₂ at 10 L/min ; cool to 38 °C ; urine output > 1-2 mL/kg/h ; gene RYR1, chromosome 19
LAST	20 % lipid emulsion 1.5 mL/kg bolus, then 0.25 mL/kg/min
Aldrete	5 parameters × 0-2 = max 10; discharge ≥ 9 ; PADSS also ≥ 9
Apfel PONV	4 factors → 10 / 20 / 40 / 60 / 80 %
Post-op fever	Wind 1-2 • Water 3-5 • Walking 4-6 • Wound 5-7 • Wonder drugs > 7 days
Wound healing	Inflammatory 0-4 d • proliferative 4-21 d • remodelling 21 d-1 y ; tensile strength 5-10 % at 1 wk, 30-50 % at 1 month, max 70-80 %
Dehiscence	POD 5-10 ; herald sign = serosanguineous ‘salmon-pink’ discharge
Suture removal	Face 3-5 d • scalp/trunk 7-10 d • limbs 10-14 d • joints/palms/soles 14 d
Wound class SSI risk	Clean < 2 % • clean-contaminated 3-10 % • contaminated 15-20 % • dirty 30-40 %
SSI definition window	30 days (90 days if an implant is present)
Ileus recovery	Small bowel 24 h → stomach 24-48 h → colon 48-72 h
Urine output	> 0.5 mL/kg/h (≈ 30 mL/h) ; maintenance fluid 4-2-1 rule / 25-30 mL/kg/day
Radiation	1 Gy = 100 rad • 1 Sv = 100 rem ; occupational 20 mSv/y (max 50 in one year) ; public 1 mSv/y ; eye 20 mSv/y ; extremities 500 mSv/y ; fetus 1 mSv (0.5 mSv/month); apron 0.25-0.5 mm Pb ; stand ≥ 2 m (6 ft) ; doubling distance → ¼ exposure
Needle-stick risk	HBV 6-30 % • HCV 1.8 % • HIV 0.3 % ; HIV PEP within 2 h (max 72 h) for 28 days ; HBIG 0.06 mL/kg within 24 h
Exposure limits	EO 1 ppm • formaldehyde 0.75 ppm • glutaraldehyde 0.05 ppm ceiling • N₂O 25 ppm • volatiles 2 ppm • methyl methacrylate 100 ppm • noise 85 dB
Lifting	NIOSH 51 lb (23 kg) ; patient handling 35 lb (16 kg)
Waste bins	Yellow anatomical/soiled/expired drugs/cytotoxic • Red recyclable plastics • White sharps • Blue glass & metallic implants; bags filled to ¾; cytotoxic incineration ≥ 1200 °C
Fatigue	≤ 12 consecutive hours direct care; ≤ 60 h/week including call; 8 h uninterrupted sleep
Drills	Fire drill ≥ twice a year ; BLS recertification every 2 years

PART F — EXAM-STYLE MCQ BANK WITH EXPLANATIONS

Position	Nerve/structure at risk	Mechanism & prevention	Physiological effect
Lateral (kidney / thoracotomy)	Dependent brachial plexus & axillary vessels, dependent common peroneal & ear, suprascapular n.	Axillary (chest) roll placed caudal to the axilla, not in it ; pillow between the legs; dependent leg flexed; head/neck in line with the spine; tape over benzoin-protected skin	V/Q mismatch — dependent lung is better perfused but less ventilated; kidney rest ↓ venous return; ↓ compliance ; post-op ‘unnatural position’ muscle pain
Jackknife / Kraske	Face, eyes, genitalia, knees; venous pooling in dependent legs	Padding at hips, pillow under pelvis and shins, arms supported	Marked ↓ venous return and ↓ FRC — poorly tolerated; ensure gradual positioning
Fracture table / traction	Perineal post → pudendal nerve injury , genital/perineal pressure necrosis; contralateral limb	Well-padded perineal post, minimum effective traction, limit traction time, protect the well leg	Blood loss, fat embolism risk in hip surgery

POSITIONING — GOLDEN RULES + MNEMONIC
Never move an anaesthetised patient without the anaesthetist's permission and direction; the **anaesthetist guards the head, airway and neck**, the **surgeon protects fractures/deformities**, the **circulator applies the padding and the safety strap**.
Position is determined by the **surgical approach + patient's physical condition + anaesthetic technique + surgeon's preference**.
The **safety strap across the mid-thighs, ~3 inches (7.5 cm) above the knees**, is the **single most important measure to prevent a fall from the table**; two fingers should slide under it.
PADDING mnemonic — Pressure points padded, **A**lignment maintained, **D**ignity/exposure minimised, **D**evelop devices in the room before the patient, IV lines & catheters protected, **N**erves free of stretch and compression, **G**entle simultaneous movement.
Compartment syndrome after lithotomy presents as **pain out of proportion, tense swollen calf, paraesthesia**; it is a **surgical emergency → fasciotomy**. Always document the ability to move limbs and any paraesthesia on arrival in PACU (your PDF's Nursing Focus III note).

B7. Safe Patient Transfer & Handling — numbers to quote

- Minimum **two persons** to transfer (your PDF); ≥ **4 persons or a mechanical lifter** for an anaesthetised, obese or unstable patient; **at least one person on each side** of the table.
- Lock the wheels** of both the gurney and the table; the gurney is placed **against** the table; a **transfer/roller board (Davis roller), slide sheet, Hoyer pad or air-assisted lateral transfer mattress** reduces the effective load.
- NIOSH revised lifting equation → maximum recommended load under ideal conditions = 51 lb (23 kg)**. For **patient handling, the recommended maximum manual lift is 35 lb (16 kg)** — above this, use assistive devices. This is the number the PDF's ‘lift team’ paragraph is really about.
- Body mechanics: **bend the knees and hips, keep the back straight, hold the load close, feet apart for a wide base, pivot with the feet — never twist the spine, push rather than pull**, lift on a count in unison.
- OSHA (2002) requires facilities to reduce ergonomic injury; **back injury is the commonest occupational injury in health care**.

- During transfer protect: **airway/ET tube, IV lines, arterial and CVP lines, epidural catheter, Foley, drains** — and the **fingers** when the table sections are raised.

B8. Electrosurgery — the physics, circuits and hazards

Fundamentals

Concept	Detail
Principle	High-frequency alternating current is concentrated at a small active electrode; tissue resistance converts it to heat → desiccation, coagulation (fulguration) or cutting (vaporisation) . It is not ‘cautery’ (true cautery = a passively heated wire; no current passes through the patient)
Frequency	300 kHz - 3 MHz (radiofrequency range) . Frequencies above ~100 kHz do not stimulate nerve or muscle (the Faradic effect is avoided) — this is why household 50/60 Hz current electrocutes but RF current does not
Cut mode	Continuous, low-voltage, high-current waveform (100 % duty cycle) → rapid vaporisation, minimal haemostasis
Coag mode	Interrupted, high-voltage, low duty cycle (~6 %) waveform → coagulum/eschar; high voltage means greater risk of capacitive coupling, insulation breakdown and sparking
Blend	Intermediate duty cycle — cutting with haemostasis
Monopolar circuit	Generator → active electrode (pencil) → patient → dispersive/return electrode (‘ground’ pad) → generator. Current traverses the whole body; the pad is essential
Bipolar circuit	Both electrodes are the two tines of the forceps ; current passes only through the tissue between them. No dispersive pad needed , minimal lateral spread, safest with pacemakers , usable in fluid, low power — used in neurosurgery, ophthalmic and microsurgery
Dispersive pad placement	Over a clean, dry, well-vascularised, hair-free muscle mass (thigh, flank), as close as practicable to the operative site, entire surface in uniform contact, away from ECG electrodes , bony prominences, scar tissue, tattoos, metallic implants and prostheses, and away from pooled fluids
Return-electrode monitoring (REM / contact-quality monitoring)	A split (dual) pad in which the generator continuously measures impedance between the two halves and deactivates itself if contact is inadequate — has virtually eliminated pad-site burns. Modern generators are isolated rather than ground-referenced
Active-electrode monitoring (AEM)	Shielded laparoscopic instruments that detect and drain stray current — prevents insulation-failure and capacitive-coupling burns

Hazards — the five classic mechanisms of electrosurgical injury

- Return-electrode (pad-site) burn** — poor/partial contact, hair, gel dried out, pad over bone or scar. Prevention: correct placement + REM.
- Alternate-site burn** — the patient's skin contacts a **metal surface** (table edge, stirrup, arm-board screw, ECG lead) which becomes an unintended return path. Your PDF states this rule twice — **no skin-to-metal contact**.
- Direct coupling** — the activated electrode touches another metal instrument (e.g. a laparoscope or retractor) which then burns adjacent tissue.
- Capacitive coupling** — current is transferred through **intact insulation** to a nearby conductor by an electrostatic field; classic setting = a **metal instrument passing through a plastic (hybrid) trocar**

Luer-lock syringes; PPE = **two pairs of chemotherapy-tested gloves (changed every 30–60 min), an impermeable back-closing gown, eye protection and an N95/respirator; never crush or split cytotoxic tablets**; label with a ‘**Cytotoxic — handle with care**’ warning; **pregnant, breast-feeding and trying-to-conceive staff should not handle them**; dispose in the **yellow cytotoxic stream**; keep a **spill kit** and document exposure.

- Chemical spill response** — **evacuate and restrict access → identify the agent from the SDS → don appropriate PPE → contain and absorb from the outside inwards with the spill-kit absorbent → neutralise/decontaminate → place waste in a labelled hazardous-waste bag → ventilate → report and document**. For a **skin/eye splash: irrigate at the eyewash station or shower for a minimum of 15 minutes** and seek occupational-health review.

PART E — THE NUMBERS SHEET (ONE PAGE, LEARN COLD)

Topic	Number to remember
Fasting	2 h clear fluids • 4 h breast milk • 6 h formula/light meal • 8 h fatty meal
Aspiration risk	Gastric volume > 25 mL (0.4 mL/kg) and pH < 2.5
Antibiotic prophylaxis	Within 60 min of incision (120 min for vancomycin); redose after 2 half-lives or > 1500 mL blood loss; stop within 24 h
Patient identifiers	At least 2 ; WHO checklist = 3 phases, 19 items
Safety strap	Across mid-thighs, ≈ 3 inches (7.5 cm) above the knees
Arm abduction	≤ 90° ; palms up; Trendelenburg tilt 30–45°
Sterile field margins	Wrapper edge unsterile = 2.5 cm (1 inch) ; unsterile person keeps 30 cm (1 ft) away; gown sterile chest to field level , sleeves to 5 cm above the elbow
Surgical scrub	First scrub of the day 3-5 min
OR environment	Temp 20–23 °C • RH 20–60 % • ≥ 15 (ideally 20–25) air changes/h • POSITIVE pressure • HEPA 99.97 % at 0.3 µm • laminar flow 300–600 ACH
Autoclave	121 °C / 15 psi / 15–20 min • 134 °C / 30 psi / 3–3.5 min • prevacuum 132–135 °C / 3–4 min • flash/IUSS 132 °C / 3 min unwrapped (10 min lumened) • prions 134 °C / 18 min
Hot air oven	160 °C / 2 h • 170 °C / 1 h • 180 °C / 30 min
Bowie-Dick	134 °C for 3.5 min in an EMPTY prevacuum chamber, daily — tests air removal
Ethylene oxide	450–1200 mg/L • 37–63 °C • RH 40–80 % • 1–6 h ; aeration 8–12 h at 50–60 °C or 7 days at room temperature; PEL 1 ppm
H ₂ O ₂ plasma	58 % H₂O₂, 45–50 °C, ≈ 1 h ; no cellulose, no liquids
Glutaraldehyde 2 %	Sterilization 10 h • HLD 20–45 min (textbook: 10 min); ceiling 0.05 ppm
Other chemicals	OPA 0.55 % — 12 min • peracetic acid — 12 min at 50–56 °C • H₂O₂ 7.5 % — 30 min HLD
Sterility	SAL = 10⁻⁶ • γ-radiation dose 25 kGy • filter 0.22 µm
Biological indicators	Steam & plasma → Geobacillus stearothermophilus • dry heat & EO → Bacillus atrophaeus • radiation → Bacillus pumilus
Electrosurgery	300 kHz - 3 MHz ; nerve stimulation avoided above ~100 kHz ; ultrasonic scalpel 55.5 kHz ; LigaSure seals up to 7 mm vessels
Surgical smoke	1 g of tissue ≈ 3–6 cigarettes ; evacuator within 2.5–5 cm (1–2 in) ; ULPA 99.999 % at 0.1 µm

Agent	Limit	Health effects & control
Halogenated volatile agents (halothane, isoflurane, sevoflurane)	NIOSH REL 2 ppm (0.5 ppm if used with N ₂ O)	Headache, fatigue, impaired mental performance, audiovisual ability and manual dexterity ; hepatotoxicity; teratogenic concern
Ethylene oxide	OSHA PEL 1 ppm (8-h TWA) ; excursion 5 ppm/15 min ; action level 0.5 ppm	Mucosal irritation, mutagen, carcinogen (leukaemia/lymphoma) , reproductive toxicity, flammable/explosive. Control: dedicated ventilated room, gas monitors and alarms, aeration cabinet, personal badges
Formaldehyde	OSHA PEL 0.75 ppm TWA; STEL 2 ppm	Eye/airway irritation, asthma, dermatitis, Group 1 human carcinogen (nasopharyngeal) — this is why routine OT fumigation is no longer advised. Handle specimens in a ventilated hood with gloves and eye protection
Glutaraldehyde	ACGIH ceiling 0.05 ppm (no safe TWA)	Occupational asthma, rhinitis, conjunctivitis, contact dermatitis, epistaxis, headache. Control: covered/closed soaking basin with local exhaust, nitrile or butyl gloves (NOT latex — it permeates), goggles, apron; thorough rinsing of instruments
Methyl methacrylate (bone cement)	OSHA PEL 100 ppm	Pungent, irritant, dermatitis, headache; cement implantation syndrome in the patient — hypotension, hypoxia, arrhythmia and cardiac arrest during cementing. Control: mix in a closed vacuum system with local exhaust , nitrile gloves, no contact with soft contact lenses (absorbs vapour)
Peracetic acid / hydrogen peroxide	Manufacturer/ACGIH limits	Irritant, corrosive at high concentration; largely safe breakdown products
Surgical smoke	No formal PEL; treated as an aerosol hazard	See §B8 — smoke evacuator with ULPA filter + high-filtration mask/N95
Noise	OSHA action level 85 dB (8-h TWA)	Saws, drills, suction, alarms, music — hearing protection and noise reduction; also impairs team communication
Compressed gases	—	Secure every cylinder upright in a stand or chained; never drop a cylinder or let one strike another; transport with the valve cap on; never oil/grease an oxygen valve; store away from heat; segregate full and empty cylinders. A cylinder that falls can become a missile

Hazard communication, cytotoxics and spills

- **Hazard Communication Standard / ‘Right to Know’:** every hazardous chemical must have a **Safety Data Sheet (SDS — formerly MSDS), in 16 standardised sections**, accessible to every employee at all times, plus **GHS labelling** with the signal words **‘Danger’** or **‘Warning’** and the nine pictograms (flame, flame over circle/oxidiser, exploding bomb, corrosion, gas cylinder, skull and crossbones, exclamation mark, health hazard, environment).
- **Hierarchy of controls (in order):** elimination → substitution → engineering controls (ventilation, closed systems, scavenging) → administrative/work-practice controls → PPE (the LAST and weakest line of defence) — a very commonly asked ranking.
- **Cytotoxic / hazardous drug handling (USP <800> principles):** prepare only in a **Class II Type B biological safety cabinet or a compounding aseptic containment isolator, inside a negatively-pressured, externally-vented room**; use **closed-system transfer devices** and

cannula during laparoscopy, or coiled cords. Worse with **high-voltage coag mode and long, thin instruments**.

- Insulation failure** — a pin-hole break in the shaft of a laparoscopic instrument delivers current **out of the surgeon's field of view** → delayed bowel perforation presenting 3–10 days post-operatively. Inspect insulation before every use.

Other electrosurgery examination points

- **Repeated requests to increase the power setting** = poor pad contact, a break in a connection, a frayed cord or a failing generator — **stop, check connections and pad; do not simply turn it up.** Replace the unit, label it, and **document the serial numbers and the reason.**
- **Keep the pencil in its holster** when not in use (direct-contact burns and drape fires). Clean the eschar off the tip with a **non-abrasive scratch pad**, not a scalpel.
- **Pacemaker / ICD:** use **bipolar** where possible; if monopolar is essential use short bursts at the lowest power, keep the current path away from the device (pad so that the pad-site line does not cross the generator), have **magnet, external defibrillator/pacer and the crash cart immediately available**, monitor ECG and pulse continuously, and consider re-programming/deactivating anti-tachycardia therapy pre-operatively. **Interrogate the device after surgery.** Also avoid MRI and diathermy near IEDs.
- **Do not use ESU or laser in an oxygen-enriched field** (mouth, oropharynx, tracheostomy, or under drapes with high-flow O₂) — pause the oxygen or reduce FiO₂ and wait for washout.
- **Advanced energy devices — Ultrasonic (Harmonic) scalpel:** mechanical vibration at **55.5 kHz**, protein denaturation by cavitation at a **lower temperature (50–100 °C)**, no electrical current through the patient, minimal smoke, seals vessels up to 5 mm. **Bipolar vessel sealing (LigaSure):** pressure + energy seals vessels up to **7 mm** with feedback-controlled cut-off. **Argon-enhanced coagulation:** argon gas jet carries current, gives a thin uniform eschar (risk of gas embolism). **Radiofrequency ablation, cryosurgery (- 60 to - 190 °C), microwave** — alternatives.
- **Surgical smoke (plume):** contains **> 150 chemicals** including benzene, formaldehyde, hydrogen cyanide, acrolein, plus viable cells, blood fragments and viral DNA (**HPV transmission to surgeons is documented**). Ablating **1 g of tissue ≈ the mutagenic exposure of 3–6 unfiltered cigarettes.** Control: a **smoke evacuator with an ULPA filter (99.999 % at 0.1 μm) held within 2.5–5 cm (1–2 inches)** of the plume, plus a **high-filtration surgical mask (or N95)** — a standard mask is *not* adequate protection.

B9. Tourniquet, Laser and Powered Equipment — quick numbers

Device	Key parameters & safety
Pneumatic tourniquet	Cuff ends overlap 3–6 inches; never at or near the elbow (radial nerve) or the mid-humerus/fibular neck. Pressure: upper limb 250–300 mmHg (or systolic + 100), lower limb 300–350 mmHg (or 2 × systolic); less in children and thin adults. Time: ≤ 60 min upper limb, ≤ 90–120 min lower limb ; if longer, deflate for 10–15 min then re-inflate. Exsanguinate with an Esmarch bandage from distal to proximal before inflation. Record site, pressure, and inflation and deflation times on the perioperative and anaesthesia records. Complications: nerve palsy, post-tourniquet syndrome , rhabdomyolysis, compartment syndrome, tourniquet pain and hypertension , and on release — ↓ BP, transient metabolic acidosis, ↑ K⁺, ↑ EtCO₂ and risk of PE. Contraindicated in sickle-cell disease, severe PVD, DVT, crush injury

Device	Key parameters & safety
Surgical laser	Light Amplification by Stimulated Emission of Radiation. Classes 1-4; all surgical lasers are Class 3B or 4 → mandatory controls. Wavelength-specific eye protection for everyone including the patient (CO ₂ 10 600 nm — clear glass/plastic suffices and cornea absorbs; Nd:YAG 1064 nm — penetrates deeply, needs green-tinted goggles; Argon 488/514 nm — amber; KTP 532 nm). Controls: warning signs on all doors, windows covered, restricted entry, key kept by the designated laser safety officer, ‘standby’ mode when not firing, foot pedal only to the surgeon, matte/non-reflective instruments, wet towels and drapes around the site, saline available, special fire-extinguisher in room, smoke (plume) evacuator, laser-safe cuffed ET tube with saline-filled cuff and no nitrous oxide, and never use in an oxygen-enriched field. Hazards: eye (retinal/corneal) injury, skin burn, fire and airway fire, plume, electrical
Powered saws/drills	Engineering clearance before first use; test before use; ensure attachments are secure and sharp; jigs and templates size-matched; let the rotating part come to a complete standstill before setting it down; deactivate the power source when not in use for an interval; irrigate to reduce thermal necrosis; face shield mandatory (bone chips, aerosol)
Pulsed lavage	Single-use; face shield for all scrubbed persons; do not submerge the handle; not near an oxygen-enriched area or flammable anaesthetics; dispose of unit and batteries as biohazard

B10. Counting Procedure — Retained Surgical Items (RSI)

The PDF gives the 9 counting rules perfectly — learn them from there. What it lacks is the epidemiology and terminology.

- **RSI / RFO (retained foreign object)** is classified by both the JC and CMS as a ‘never event’ / sentinel event.
- **Commonest retained item = a surgical sponge/gauze** (≈ 50–70 %), then instruments (retractors, malleables) and needles. Commonest site = **abdomen/pelvis** (then vagina and thorax).
- A retained gauze surrounded by a foreign-body granuloma is called a **gossypiboma / textiloma / cottonoid granuloma** — presents weeks to years later with a mass, pain, sinus, abscess, obstruction or fistula, and is a classic **res ipsa loquitur** claim.
- **Risk factors:** emergency surgery, **unexpected intra-operative change in procedure**, high BMI, more than one surgical team/procedure, multiple staff changes, incorrect or omitted counts, and long operations.
- **Counts are performed:** (1) before the procedure (baseline), (2) when **new items are added**, (3) **before closure of a cavity within a cavity** (uterus, bladder, bowel), (4) when **wound (fascia) closure begins**, (5) at **skin closure / end of procedure**, and (6) at any **change of scrub or circulating personnel**.
- **Adjunct technologies:** all sponges used in a wound must be **radio-opaque (X-ray detectable)**; **bar-coded sponge counting systems**; **radiofrequency (RF) tag detection wands/mats** that scan the patient before closure; intra-operative radiography.
- **If the count is incorrect:** repeat the count → **inform the surgeon immediately** → surgeon explores the wound → if still missing, **intra-operative X-ray before the patient leaves the OR** → notify the OR supervisor → **document the discrepancy, the actions and the outcome, and file an incident report**. **Nothing (linen, trash, instruments, sponges) leaves the room until the final count is correct.**
- **X-ray-detectable sponges are never used as dressings**, and dressing sponges are never used in the wound.

Colour	Category	Contents	Treatment / disposal
RED	Contaminated recyclable waste (plastics)	IV tubing and sets, catheters, urine bags, syringes without needles , vacutainers without needles, gloves	Autoclaving / microwaving / hydroclaving, then shredding → sent to a registered recycler. No chemical pre-treatment before recycling; no landfilling
WHITE (translucent, puncture-proof, leak-proof)	Waste SHARPS including metals	Needles, syringes with fixed needles , needles from needle-tip cutters, scalpels, blades, any contaminated sharp object that may cause puncture	Autoclaving/dry-heat sterilisation followed by shredding or mutilation; final disposal by encapsulation in a concrete/metal sharps pit, or sent to an iron foundry / sanitary landfill
BLUE (cardboard box with blue marking / puncture-proof of container)	Glassware and metallic body implants	Broken or discarded and contaminated glass including medicine vials and ampoules (except those contaminated with cytotoxic drugs — those go in yellow), metallic implants	Disinfection (1 % hypochlorite) or autoclaving/microwaving/hydroclaving, then sent for recycling

- **Golden rule: segregate at the point of generation, by the person generating the waste — never sort waste later.** Bags are filled to $\frac{3}{4}$ **capacity**, tied (never stapled), labelled with the **biohazard/cytotoxic symbol** and the department/date, and stored no longer than **48 hours**.
- **Never recap, bend or break needles by hand; use a needle-tip cutter or destroyer at the point of use (the needle goes to the WHITE container).**
- **Liquid waste** (laboratory, washing, cleaning): pre-treat with disinfectant, then discharge to a sewer/ETP as per the Effluent standards.
- **Mercury** (broken thermometer/sphygmomanometer): **do not vacuum**; collect droplets with a syringe/adhesive tape or a mercury spill kit, place in a sealed container of water, dispose as hazardous waste — mercury is a neurotoxin. Mercury devices should be phased out.
- **Records, training and reporting:** annual report to the Pollution Control Board, maintenance of a waste register, **annual health check-up and immunisation (hepatitis B, tetanus) of waste handlers**, provision of PPE, and reporting of any **major accident** within 24 h.
- **Radioactive waste** is governed separately by the **Atomic Energy Regulatory Board (AERB)** — decay-in-storage in a shielded room until background level, then normal disposal.

D4. Chemical & Physical Hazards — exposure limits and control

Occupational exposure limits worth memorising

Agent	Limit	Health effects & control
Nitrous oxide	NIOSH REL 25 ppm (TWA over the procedure)	Bone-marrow/megaloblastic change (methionine synthase inhibition), neuropathy, ↑ spontaneous abortion and reduced fertility , addiction. Control: scavenging system, no leaks, tight-fitting masks, ≥ 15–25 air changes/h

The most contaminated items come off first (gloves), and the respirator comes off last because it protects you until you have left the room. Hand hygiene both before donning and immediately after doffing, and at any point where PPE is breached.

Sharps injury and post-exposure management

Item	Detail
Transmission risk from a single percutaneous needle-stick from an infected source	Hepatitis B ≈ 6-30 % (up to 30 % if HBeAg-positive; the highest risk) • Hepatitis C ≈ 1.8 % (range 0-7 %) • HIV ≈ 0.3 % (1 in 300) ; mucous-membrane exposure to HIV ≈ 0.09 %
Immediate first aid	Do NOT squeeze or suck the wound and do NOT use caustic agents (bleach, alcohol, iodine) inside the wound. Wash with soap and running water (irrigate mucous membranes/eye copiously with water or saline for 10-15 min, remove contact lenses), allow it to bleed freely, dry and cover
Then	Report immediately to the designated officer/occupational health → assess the exposure (device, depth, volume, source status) → baseline serology of the exposed person and, with consent, the source → start PEP → counselling → follow-up serology at 6 weeks, 3 months and 6 months → document and file an incident report
HIV PEP	Start as early as possible — ideally within 2 hours, and no later than 72 hours; 3-drug regimen (e.g. tenofovir + lamivudine/emtricitabine + dolutegravir) for 28 days ; the exposed person should use barrier contraception and not donate blood during follow-up
Hepatitis B PEP	If the exposed person is unvaccinated or a non-responder → hepatitis B immunoglobulin (HBIG) 0.06 mL/kg IM within 24 hours (up to 7 days) PLUS start the vaccine series. A vaccinated responder (anti-HBs ≥ 10 mIU/mL) needs no prophylaxis. Hepatitis B vaccination is mandatory for all health-care workers — 3 doses at 0, 1 and 6 months, with anti-HBs titre checked 1-2 months after the last dose
Hepatitis C	No vaccine and no PEP — monitor with HCV RNA at 3-6 weeks and antibody at 4-6 months; treat early infection with direct-acting antivirals
Prevention (hierarchy of controls)	Elimination/substitution (needleless systems) → engineering controls (retractable and sheathed needles, blunt suture needles, puncture-proof sharps container filled to only ¾, scalpel-blade removers) → administrative/work-practice controls (never recap by hand — if unavoidable, one-handed ‘scoop’ technique; never pass a sharp hand-to-hand — use a neutral zone / hands-free ‘no-touch’ technique with a magnetic pad or basin; mount the blade with a needle holder; announce sharps) → PPE (double gloving reduces inner-glove perforation by ≈ 70-80 %, blunt needles, face shield)

Biomedical waste segregation — Biomedical Waste Management Rules 2016 (India)

Colour	Category	Contents	Treatment / disposal
YELLOW	Human & animal anatomical waste; soiled waste; expired/discarded medicines; chemical waste; chemical liquid waste; discarded linen contaminated with blood/body fluids; microbiology, biotechnology and other clinical laboratory waste	Body parts, placenta, tissue, organs, dressings, cotton, blood-soaked items, plaster casts, expired drugs, cytotoxic drugs and vials (in a separate yellow ‘cytotoxic’ bag/container)	Incineration / plasma pyrolysis / deep burial (deep burial only in towns < 500 000 population). Cytotoxic waste → incineration at ≥ 1200 °C or return to the manufacturer

B11. Operative / Perioperative Records — beyond the forms

Specimen handling — a favourite MCQ that the PDF omits

Specimen type	Handling
Routine histopathology	10 % neutral buffered formalin , volume ≈ 10× the specimen ; container labelled with patient name, ID, date, site/side and specimen type; requisition matched; label read aloud during ‘sign out’
Frozen section	Fresh, in saline-moistened gauze — NEVER in formalin (formalin makes further study and immunohistochemistry impossible); send immediately
Culture / microbiology	Sterile container, no fixative ; anaerobic swab or syringe for anaerobes; transport immediately; never in formalin
Cytology / fluid	Fresh or with cytology fixative as per laboratory
Stones/calculi	Dry container, no fixative
Amputated limb / tissue for the patient	Per state law, hospital policy and cultural/religious wishes; documented
Lost specimen	Treated as a serious incident — irreplaceable; incident report mandatory
Implants	Record type, manufacturer, size, serial/lot number in the intra-operative record and the implant log (traceability for recalls)

What makes the record legally sound

- **JC-mandated minimum content of the operative record:** surgeon's name (and assistants), **pre-operative diagnosis**, procedure performed, **post-operative diagnosis**, findings, **specimens removed (type and number)**, estimated blood loss, and the names of all team members providing intra-operative care.
- Also document: the **time out**, counts and their outcome, skin condition **before and after** prep and the prep solution, position and positioning aids, **ESU pad site, serial number and settings**, tourniquet data, drains/catheters/packing, medications and irrigations (name, strength, dose, route, time, who gave them), implants, intake/output, transfusion checks, and any **unusual event with the actions taken**.
- **Rules of documentation:** contemporaneous, legible, factual and objective (no opinion or blame), signed and timed; **corrections by a single line through with the author's initials, date and time — never erase, overwrite or use correction fluid**; no blank spaces left; late entries labelled as such.
- **Incident (occurrence) report** — an internal risk-management document. Record only the facts; do **not** photocopy it into the chart or reference it in the clinical notes in many jurisdictions (the PDF's practice of filing it in the chart follows its own institutional convention — answer as per your textbook if asked). File one for: incorrect count, medication error, burn, fall, retained item, equipment failure, needle-stick, specimen loss, unplanned re-operation and **death in the OR**.
- **Retention:** medical records are typically retained for **3 years** for outpatient and **10 years** for in-patient/medico-legal records in Indian practice (MCI/NMC guidance; longer for minors and medico-legal cases) — follow institutional policy.
- **Confidentiality** — patient information is disclosed only on a need-to-know basis; consent needed for photography, and images in which the patient is identifiable.

B12. Sterilization & Disinfection — the full science (highest-yield section)

Definitions — get these exactly right

Term	Definition
Sterilization	Complete destruction or removal of all forms of microbial life, including bacterial and fungal spores . It is an absolute term — an item is either sterile or it is not
Disinfection	Destruction of pathogenic organisms but not necessarily spores ; used on inanimate objects
Antisepsis	Destruction/inhibition of organisms on living tissue (skin, mucosa)
Sanitisation	Reduction of microbial load to a level judged safe by public-health standards
Decontamination	Physical/chemical removal or inactivation of contaminants so that an item is safe to handle — always the FIRST step, before disinfection or sterilization
Cleaning	Physical removal of soil and organic matter (cannot sterilize a dirty instrument — organic matter blocks the sterilant)
Asepsis	Absence of pathogenic organisms; medical asepsis = clean technique (reduce numbers), surgical asepsis = sterile technique (absence of all organisms)
Bactericidal / -static	Kills / merely inhibits multiplication
Sporicidal	Kills spores — the property that distinguishes a sterilant from a disinfectant
SAL (Sterility Assurance Level)	10⁻⁶ — probability of $\leq$ one viable organism in 10 ⁶ sterilized items. The internationally accepted definition of ‘sterile’
D-value	Time (or radiation dose) needed to reduce the microbial population by 1 log₁₀ (90 %) at a stated condition
Bioburden	Number and resistance of microbes on an item before sterilization

Spaulding classification — how the level of processing is decided

Category	Definition	Examples	Required processing
Critical	Enter sterile tissue or the vascular system	Surgical instruments, implants, needles, cardiac & urinary catheters, arthroscopes, laparoscopes	Sterilization
Semi-critical	Contact intact mucous membranes or non-intact skin	Flexible endoscopes (GI, bronchoscope), laryngoscope blades, ET tubes, respiratory & anaesthesia equipment, vaginal probes, thermometers	High-level disinfection (sterilization preferred where feasible)
Non-critical	Contact intact skin only	BP cuffs, stethoscopes, ECG electrodes, bedpans, OR table, floors, walls, furniture	Low- or intermediate-level disinfection / cleaning

Resistance of organisms to sterilization / disinfection (most → least resistant)

Prions > Bacterial spores (C. difficile, Bacillus, Geobacillus) > Coccidia (Cryptosporidium) > Mycobacteria > Non-lipid/small viruses (polio, norovirus, HAV) > Fungi > Vegetative bacteria > Lipid/medium-sized viruses (HIV, HBV, herpes, influenza) — hence a sporicidal agent covers everything below spores, and HIV/HBV are among the easiest to kill (a common MCQ inversion).

Effect	Nature	Examples
Stochastic	No threshold ; the probability (not the severity) increases with dose; assumed linear-no-threshold	Carcinogenesis (leukaemia, thyroid, breast, lung) and heritable genetic mutation

Most radiosensitive tissues (rapidly dividing): bone marrow/lymphoid tissue, gonads (spermatogonia), intestinal crypt epithelium, skin basal layer, lens, fetus — the fetus is most sensitive during organogenesis, weeks 2-8. Least sensitive: nerve, muscle and bone in the adult.

D3. Infection Control — precautions, exposure management, and waste

Standard vs transmission-based precautions

Category	Applies to	Requirements	Examples
Standard Precautions (CDC 1996; incorporate the earlier Universal Precautions and Body Substance Isolation)	EVERY patient, every time, regardless of diagnosis — all blood, all body fluids, secretions and excretions except sweat , non-intact skin and mucous membranes	Hand hygiene • gloves • gown • mask/eye protection or face shield when splash is likely • safe injection practices and sharps handling • respiratory hygiene/cough etiquette • safe handling of contaminated equipment, linen and waste • environmental cleaning • patient placement	Universal application in the OR
Contact Precautions	Direct or indirect contact transmission	Standard + gown and gloves for all contact , dedicated equipment, single room preferred	MRSA, VRE, ESBL/CRE, C. difficile (soap and water — alcohol does not kill spores ; hypochlorite for the environment), scabies, RSV, hepatitis A
Droplet Precautions	Large droplets (> 5 µm) that travel ≈ 3-6 feet (1-2 m)	Standard + surgical mask within 1-2 m , single room, mask on the patient during transport	Influenza, pertussis, mumps, rubella, Neisseria meningitidis, diphtheria, group A streptococcus, adenovirus
Airborne Precautions	Small droplet nuclei ($\leq$ 5 µm) that remain suspended and travel long distances	Standard + N95 (or higher) respirator, fit-tested , and an Airborne Infection Isolation Room — NEGATIVE pressure with $\geq$ 12 air changes/hour, exhaust to outside or HEPA-filtered , door kept closed	Pulmonary/laryngeal tuberculosis, measles (rubeola), varicella/disseminated zoster, SARS-CoV-2 during aerosol-generating procedures

PPE sequence — rote-learn the order

Donning (put on)	Doffing (take off)
1 Hand hygiene → 2 Gown → 3 Mask or respirator (fit-check the N95) → 4 Goggles / face shield → 5 Gloves	1 Gloves → 2 Goggles/face shield → 3 Gown → 4 Mask/respirator (remove outside the room; touch only the ties/elastic, never the front) → 5 Hand hygiene

Category	Limit
Occupational — skin, hands and feet (extremities)	500 mSv/year
Pregnant worker (after declaration of pregnancy)	≤ 1 mSv to the fetus/abdominal surface for the remainder of the pregnancy (US NCRP: 0.5 mSv per month)
Members of the public	1 mSv per year
Trainees / apprentices aged 16–18	6 mSv per year

Protection — the three cardinal principles + practical rules

Principle	Application
1. Time	Minimise fluoroscopy screening time; use pulsed fluoroscopy, last-image hold and collimation ; the operator, not the assistant, should control the pedal; record total screening time on the perioperative record
2. Distance — the INVERSE SQUARE LAW	Intensity ∝ 1/(distance)². Doubling the distance reduces exposure to one quarter (25 %); tripling it to one ninth. Practical rule: stand at least 6 feet (2 m) from the source, or leave the room. In lateral fluoroscopy stand on the image-intensifier side, away from the X-ray tube , because scatter is highest on the tube side
3. Shielding	Lead apron 0.25-0.5 mm lead equivalence (0.5 mm attenuates ≈ 95–99 % of the primary beam at 100 kVp); thyroid collar, lead glasses (0.75 mm Pb equivalent lenses), lead gloves for holding cassettes, mobile lead screens and ceiling-suspended shields. Aprons must be hung, never folded , and radiographically screened annually for cracks . Wrap-around aprons for procedures where you turn your back to the source
ALARA	As Low As Reasonably Achievable — the governing philosophy; justification, optimisation and dose limitation (the three ICRP pillars)
Monitoring	Personal dosimeter worn at chest/collar level OUTSIDE the lead apron (a second one inside for pregnant staff at waist level). Types: TLD (thermoluminescent, LiF — most common, read monthly/quarterly), film badge, OSL, and a pocket dosimeter/electronic personal dosimeter for real-time reading. Records are kept for the worker's lifetime
Personnel rules	All non-essential staff leave the room or stand behind a lead screen; scrubbed staff wear a lead apron donned BEFORE scrubbing; never hand-hold the cassette or the patient if it can be avoided — use lead gloves and a holder if unavoidable; announce ‘X-ray!’ before every exposure; pregnant staff should be excluded from fluoroscopic cases
Patient rules	Ask every woman of child-bearing age whether she is or could be pregnant (the ‘10-day rule’ for non-urgent radiography); shield the gonads/reproductive organs and thyroid when the beam cannot be collimated; collimate to the area of interest; use the lowest diagnostic exposure; document fluoroscopy time and dose
Radioactive implants (bra chytherapy)	Handle sources with long forceps, never with fingers ; keep a lead pot/container in the room; limit staff time in the room and rotate staff; restrict visitors and exclude pregnant staff and children; survey the room and linen with a Geiger-Müller counter after removal; specific disposal and incident procedures for a lost source

Biological effects — the distinction examiners test

Effect	Nature	Examples
Deterministic (non-stochastic)	Has a threshold dose ; severity increases with dose	Skin erythema (≈ 2 Gy) and desquamation, epilation, cataract , sterility, bone-marrow suppression, acute radiation syndrome, fetal malformation

PRIONS — the exception to every rule (CJD)
Prions resist routine autoclaving, EO, formaldehyde, alcohol and plasma sterilization.
Recommended: **single-use instruments where possible**, otherwise **1 N NaOH (or 1 % NaOCl) for 1 h + steam autoclaving at 134 °C for 18 minutes** (prevacuum), or 121 °C for 60 min after NaOH immersion.
Never flash-sterilize; do not reuse in neurosurgery.

1. Moist heat / saturated steam under pressure (autoclave) — the parameters

Mechanism: denaturation and coagulation of proteins. Moist heat is far more efficient than dry heat because latent heat of vaporisation is released when steam condenses on the item. It is the method of choice — fastest, cheapest, most reliable, non-toxic — for everything that can tolerate heat and moisture.

Cycle / type	Temperatu re	Pressure	Hold time	Use
Gravity displacement (downward)	121 °C (250 °F)	15 psi (≈ 1 atm / 103 kPa)	15–20 min (wrapped packs 30 min)	Standard general load, glassware, media, rubber
Gravity displacement — high temp	134 °C (273 °F)	30 psi (≈ 2 atm)	3–3.5 min	Rapid cycles
Prevacuum (high-vacuum, pulsed)	132–135 °C	≈ 30 psi	3–4 min (+ drying 20–30 min)	Wrapped instruments, trays, textiles, lumened items — air is actively evacuated so steam penetrates
Flash / IUSS (Immediate-Use Steam Sterilization)	132 °C (270 °F)	≈ 27–30 psi	3 min unwrapped metal only; 10 min for porous, lumened or complex items	Only for urgently needed, unwrapped single items — e.g. a dropped instrument. Not for implants, not for convenience, not for whole sets
Prions (CJD)	134 °C	—	18 min prevacuum after NaOH	Neurosurgical instruments

- **Air is the enemy of the autoclave** — trapped air lowers the temperature at a given pressure and prevents steam contact. Hence: **do not overload, do not wrap tightly, place trays with mesh bottoms, position packs on edge with space between them, and never place a bowl/basin so that it traps air.**
- **Bowie-Dick test** = a **daily test of air removal / steam penetration in a PREVACUUM sterilizer**, run in an **empty chamber** at 134 °C for 3.5 min; a uniform colour change on the test sheet means no air pocket. It is a **Class 2 chemical indicator** and it does **not** prove sterility.
- **Vacuum leak test** weekly; **biological indicator at least weekly (daily preferred) and with EVERY implant load** — implant loads should be quarantined until the BI result is available.
- **Cannot be autoclaved:** sharp cutting edges (blunted), oils, greases, powders, petroleum-based ointments, anhydrous materials, plastics with low melting point, endoscopes with lenses, and heat-labile electronics.
- Drying is part of the cycle — a **wet pack is a contaminated pack (strike-through)**; do not touch or move packs until cool and dry.

2. Dry heat (hot air oven / incineration / flaming) — absent from your PDF

Mechanism: oxidation and protein denaturation — needs much higher temperature and longer time than moist heat.

Temperature	Holding time
160 °C	2 hours (the most commonly asked pair)
170 °C	1 hour
180 °C	30 minutes
150 °C	2.5 hours

- **Uses:** glassware, metal instruments (**sharp**s — **does not blunt or corrode**), oils, greases, powders, anhydrous fats, glass syringes, all-glass items.
- **Not for:** rubber, plastics, fabrics, most surgical linen.
- Load loosely to allow air circulation; **allow to cool before opening** (glass cracks, and hot air rushing in draws contaminated air).
- **Other dry-heat methods:** **incineration** (final disposal of contaminated waste and anatomical parts, ≥ 1100–1200 °C), **red-heat flaming** (inoculation loops), **glass-bead steriliser** (250 °C, 20–40 s — dental).
- **Biological indicator:** **Bacillus atrophaeus** (formerly B. subtilis var. niger).

3. Ethylene oxide (EO) gas chemical sterilization

Parameter	Value
Mechanism	Alkylation of DNA, RNA and proteins
The 4 critical variables	Gas concentration (450-1200 mg/L) • Temperature (37-63 °C) • Relative humidity (40-80 %) • Exposure time (1-6 h; your PDF quotes 3-7 h cycles) — all four must be met
Aeration ('degassing')	Mechanical aerator: 8-12 h at 50-60 °C (122-140 °F), air changed ≥ 4 times/h. Ambient room: 7 days at 18-22 °C — mandatory, because residual EO causes chemical burns and haemolysis
Uses	Heat- and moisture-sensitive items — flexible endoscopes, cardiac catheters, plastics, electrical/electronic equipment, prosthetic valves, pacemakers, respiratory equipment, delicate optics
Rule	Anything that can withstand steam should NOT be gas-sterilized
Hazards	Toxic, irritant, mutagenic, teratogenic, carcinogenic (leukaemia, lymphoma), flammable and explosive (hence mixed with CO ₂ or HCFC diluents); OSHA PEL 1 ppm (8-h TWA) , excursion 5 ppm/15 min, action level 0.5 ppm
Biological indicator	Bacillus atrophaeus
Limitation	Cannot penetrate anything that absorbs it ; must be scrupulously dry and clean; long turnaround

4. Low-temperature hydrogen peroxide gas plasma (STERRAD)

- **58 % hydrogen peroxide (1.8 mL)** is vaporised under vacuum, then excited by **radiofrequency energy into a plasma** of free radicals and charged particles → kills bacteria, fungi, viruses **and spores**.
- **Temperature 45-50 °C (≈ 122 °F); cycle 28-75 min (≈ 1 h); by-products = water vapour + oxygen only** → **NO aeration or cooling needed, no toxic residue**.
- Ideal for: metals and non-metals, **fibre-optic cables, endoscopes, microsurgical and electrical equipment, glass, ceramics**.

PART D — OT SAFETY FOR THE MEDICAL ASSISTANT / PHARMACIST (GAP-FILL)

D1. In-Service Education, Ergonomics & Fatigue — only the missing numbers

- **Orientation vs in-service education vs continuing education:** *orientation* = initial familiarisation with policies, procedures and the physical environment (AORN recommends a **mentor / buddy system**); *in-service* = **ongoing, employer-provided, job-specific** education (perioperative departments hold it **monthly**); *continuing education* = professional development beyond the employer, usually credit-based.
- **Mandatory annual training topics:** fire safety and evacuation, **fire drills at least twice a year (unannounced)**, disaster drill (at least annually), infection control and hand hygiene, biomedical waste handling, blood-borne pathogen/needle-stick protocol, radiation safety, laser safety, electrical safety, hazard communication (SDS), CPR/BLS recertification (**every 2 years**), lifting/ergonomics, and **competency validation for every new device before use**.
- **NIOSH lifting limit = 51 lb (23 kg)** under ideal conditions; **recommended maximum manual patient lift = 35 lb (16 kg)** — beyond that, use a mechanical aid or a lift team.
- **Fatigue:** AORN recommends **no more than 12 consecutive hours** of direct patient care and **≤ 60 hours per week including on-call**, with **8 hours of uninterrupted sleep**. Being awake **17-19 hours** impairs performance about as much as a blood alcohol of 0.05 %, and 24 h ≈ 0.10 %. **Working when judgement is impaired is an ethical breach, not just a personal risk**.
- **Other occupational hazards worth a line:** standing for long periods (varicose veins — use anti-fatigue mats and compression stockings), repetitive strain and awkward posture during laparoscopy and microsurgery, **noise > 85 dB** from saws/drills/suction (hearing protection), sharps injury, latex, psychological stress and burnout, and **second-victim** support after an adverse event.

D2. Radiation Safety — the quantitative detail your PDF lacks

Units

Quantity	SI unit	Old unit	Conversion / meaning
Exposure	Coulomb/kg	Roentgen (R)	Ionisation produced in air
Absorbed dose	Gray (Gy) = 1 J/kg	rad	1 Gy = 100 rad — energy deposited in tissue
Equivalent dose (absorbed dose × radiation weighting factor)	Sievert (Sv)	rem	1 Sv = 100 rem; 1 mSv = 100 mrem — the unit used for all dose limits and personnel monitoring
Effective dose	Sievert (Sv)	rem	Equivalent dose weighted for tissue sensitivity — whole-body risk
Activity	Becquerel (Bq) = 1 disintegration/s	Curie (Ci)	1 Ci = 3.7 × 10 ¹⁰ Bq

Dose limits (ICRP / AERB — India)

Category	Limit
Occupational — whole body	20 mSv per year averaged over 5 consecutive years (100 mSv/5 y), with a maximum of 50 mSv in any single year
Occupational — lens of the eye	20 mSv/year (ICRP 2011 revision; formerly 150 mSv)

C10. Discharge, Counselling and the Pharmacist's Role

- **Discharge criteria (day-care/ambulatory):** stable vital signs for ≥ 1 h, oriented, **PADSS ≥ 9**, pain and PONV controlled on oral medication, able to take oral fluids, minimal bleeding, able to ambulate/dress with minimal help, **a responsible escort and transport**, and **written** instructions with an emergency contact number.
- **Counsel the patient not to drive, operate machinery, sign legal documents, drink alcohol or care for a child unaided for at least 24 hours** after general anaesthesia or sedation.
- **Written discharge information must cover:** wound and dressing care, when and how to shower/bathe, suture or staple removal date, activity and lifting restrictions, driving and return-to-work advice, diet, **red-flag symptoms requiring urgent review — fever, increasing pain, spreading redness, purulent or foul discharge, wound gaping, bleeding, calf pain or swelling, chest pain or breathlessness, persistent vomiting, inability to pass urine, no bowel movement — follow-up appointment, and drain/catheter care if going home with one.**
- **Medication reconciliation (a core pharmacist deliverable):** compare the pre-admission list with the discharge list; restart the drugs that were withheld (**metformin, ACE-I/ARB, anticoagulants — with a clear restart date and dose**); stop the drugs that were only for the operation (**prophylactic antibiotics, PPIs, anti-emetics**); check for **therapeutic duplication (paracetamol in combination analgesics), interactions (NSAID + anticoagulant, tramadol + SSRI → serotonin syndrome), renal-dose adjustments and allergies**; counsel on **opioid tapering, laxative co-prescription, the finite duration of the analgesic course, and safe disposal of leftover opioids**; provide a **VTE-prophylaxis continuation plan** and injection technique training for LMWH; **document the counselling.**
- **Also the pharmacist's OT-related duties:** maintaining the OT drug tray and crash cart (stock, expiry, temperature and cold-chain logs, seal checks), **narcotic/controlled-substance accounting and register (NDPS)**, sterile preparation and dilution of injectables, ensuring **all syringes and containers on the sterile field are labelled with drug, strength and concentration**, segregating **look-alike/sound-alike and high-alert drugs** (concentrated potassium chloride, insulin, heparin, neuromuscular blockers), reporting and analysing medication errors and **adverse drug reactions (pharmacovigilance)**, and in-service education on new drugs and devices.

MEDICATION SAFETY IN THE OR — the pharmacist's exam angle

Extend the PDF's 'five rights' to the **modern eight rights**: right **patient, drug, dose, route, time**, plus right **documentation, reason/indication and response**; add the right to **refuse** and patient **education**.

Every medication and solution on the sterile field must be labelled immediately with name, strength and concentration — **unlabelled medications are ALWAYS discarded**; **all vials, syringes and containers stay in the room until the end of the case**; medications and containers are shown to the **relief person** at hand-over; calculations are **independently double-checked by a second licensed professional**.

High-alert medications in the OR: concentrated **potassium chloride** (never as a bolus), insulin, heparin, neuromuscular blockers (**'paralysing agent — warning' auxiliary label mandatory**), concentrated adrenaline, local anaesthetics, oxytocin, and any **look-alike/sound-alike pair** (e.g. ephedrine/epinephrine, heparin/insulin).

Constraints/forcing functions recommended to reduce error: **minimise the number of calculations required, stock ready-to-use dose-specific concentrations, standardise concentrations, use pre-filled syringes, bar-code verification, and regular competency assessment.**

- **Cannot be used for cellulose-containing items — paper, linen, cotton, towels, gauze, dressings — or for liquids, powders, or long narrow lumens**, because cellulose absorbs the sterilant. High capital cost.
- **Biological indicator: Geobacillus stearothermophilus.**

5. Liquid chemical sterilization / high-level disinfection

Agent	Sterilization	High-level disinfection	Notes
Glutaraldehyde 2 % (activated, alkaline — Cidex)	10 hours immersion	20-45 min at 20-25 °C (your textbook quotes 10 min — quote your textbook if the question is from it, but CDC says 20-45 min)	Non-corrosive to lenses/metal → endoscopes. Rinse thoroughly in sterile distilled water. Activated solution loses potency → test with a test strip before each use , reuse life 14-28 d. Wear polyethylene or double gloves + eye protection ; irritant, asthma/dermatitis, ACGIH ceiling 0.05 ppm
Ortho-phthalaldehyde (OPA) 0.55 %	—	12 min at 20 °C	Faster, less irritant than glutaraldehyde; stains skin/clothing grey
Peracetic acid 0.2 % (Steris System 1 / Steris 20)	12 min at 50-56 °C	—	Single-use, automatic dilution, effective despite organic soil , low surface tension → reaches lumens; decomposes to acetic acid + water + oxygen (environmentally safe); corrosive when concentrated
Hydrogen peroxide 7.5 %	6 hours	30 min	Non-toxic breakdown products
Formaldehyde (8 % + 70 % alcohol)	—	High level	Carcinogen, pungent — largely abandoned

6. Radiation and filtration sterilization

Method	Detail
Ionising (γ) radiation	Cobalt-60 source; dose 25 kGy (2.5 Mrad) . Cold, industrial-scale method for pre-packed single-use disposables — syringes, needles, gloves, catheters, sutures, blades, prosthetic heart valves, bone grafts. High penetration, no residue. Not used in hospitals (cost, shielding). BI = Bacillus pumilus
Non-ionising (UV-C, 254 nm)	Surface and air disinfection only — poor penetration (does not pass through glass, paper, water films). Used in laminar-flow hoods, operating-room air, biosafety cabinets. Hazard: keratoconjunctivitis and erythema — never expose skin or eyes; lamps must be off when the room is occupied. It is a disinfectant, not a steriliser
Filtration	0.22 μm (bacteria-retaining) membrane filters for heat-labile liquids — serum, antibiotic and vaccine solutions, IV fluids, culture media. Removes but does not kill; does not retain viruses or mycoplasma (needs 0.1 μm). HEPA filtration of air = the OR/laminar-flow application (SB2)

7. Sterilization monitoring — the three-indicator system

Type	What it is	What it proves	Examples
Physical / mechanical	Gauges, digital printout, chart recorder, cycle log, time-temperature-pressure graph	That the machine achieved the parameters; read and signed for every cycle	Sterilizer printout, thermocouple
Chemical (ISO 11140 classes 1-6)	Heat/chemical-sensitive dyes on tape, strips or cards	Exposure to the process — NOT sterility	Class 1 process indicator (autoclave tape on the outside of the pack — only distinguishes processed from unprocessed) • Class 2 specific-test indicator (Bowie-Dick) • Class 3 single-variable • Class 4 multi-variable • Class 5 integrating indicator (responds to all critical variables, correlates with the BI) • Class 6 emulating indicator (cycle-verification, most stringent). An internal chemical indicator goes in the centre of every pack
Biological (BI) — the GOLD STANDARD	A calibrated preparation of highly resistant bacterial spores processed with the load and then incubated	Actual sterilizing efficacy — the only indicator that proves the sterilization process killed spores	Steam & H₂O₂ plasma → Geobacillus stearothermophilus (thermophile, incubate 55-60 °C) • Dry heat & ethylene oxide → Bacillus atrophaeus (incubate 35-37 °C) • γ-radiation → Bacillus pumilus . Frequency: at least weekly, daily preferred, and with every implant load ; rapid-readout BIs give results in 1-3 h

PACKAGING, STORAGE & SHELF-LIFE — commonly asked
 Packaging must **permit sterilant penetration, provide a microbial barrier, allow aseptic opening and resist tearing/moisture**: muslin (double thickness, 140 thread count), non-woven polypropylene, crepe paper, paper-plastic peel pouches (‘see-through’), rigid containers with filters. **Never use closed metal containers or aluminium foil for steam.**
Shelf-life is now EVENT-RELATED, not TIME-RELATED: a pack remains sterile **until an event compromises it** — the wrapper becomes wet, torn, punctured or soiled, the seal is broken, it is dropped, or it is handled excessively. (The old “sterile for 30 days” rule is obsolete.)
 Store in a **clean, dry, dust-free, closed cupboard, 20-23 °C, humidity 30-60 %, at least 20-25 cm (8-10 in) above the floor, 45 cm (18 in) below the ceiling/sprinkler and 5 cm (2 in) from outside walls**; stock rotated **FIFO (first-in, first-out)**.
Sequence of instrument processing: point-of-use pre-cleaning (enzymatic foam spray in the OR) → transport in a closed container → **decontamination / washer-disinfector** → **ultrasonic cleaner** (cavitation removes soil from box locks, serrations, lumens) → rinse and dry → inspect, lubricate and assemble (**box locks and hinged instruments OPEN/unlocked**) → pack with internal CI → sterilize → cool/dry → store. **Never sterilize a soiled instrument.**

REUSE OF ‘SINGLE-USE’ DEVICES — the PDF's controversy, summarised
 Reprocessing a device labelled **single-use** transfers the **manufacturer's legal liability to the reprocessor** (the hospital or a third-party reprocessor), who must then meet the regulator's validation requirements (FDA rule, August 2000).
 Risks: **retained bioburden in lumens, endotoxin/pyrogen reaction, material degradation and functional failure, residual sterilant toxicity, and prion transmission.**
 AORN's position: **patient safety is the paramount consideration**; opened-but-unused items are less problematic than used items. In India, reuse must comply with hospital policy and the Medical Devices Rules — always answer that reuse is **acceptable only if validated and permitted by the manufacturer/regulator**.

C8. Fluids, Electrolytes, Nutrition and the GI Tract

- **Maintenance fluid (4-2-1 rule)**: 4 mL/kg/h for the first 10 kg + 2 mL/kg/h for the next 10 kg + 1 mL/kg/h for each kg above 20 — or roughly **25-30 mL/kg/day** with Na⁺ 1 mmol/kg/day and K⁺ 1 mmol/kg/day; add replacement of measured losses (drains, NG aspirate, urine) and third-space losses.
- **Monitor**: hourly urine output (**target > 0.5 mL/kg/h**), pulse, BP, capillary refill, JVP, daily weight, fluid balance chart, electrolytes, urea/creatinine, lactate.
- **Post-operative fluid retention (the ‘stress response’)**: ADH and aldosterone rise → salt and water retention, hyponatraemia and oliguria in the first 24-48 h — **oliguria after surgery is more often hypovolaemia than renal failure, but beware of over-transfusing hypotonic fluid → dilutional hyponatraemia (SIADH)**. Also expect **hyperglycaemia, hypokalaemia, negative nitrogen balance and a catabolic state**.
- **Post-operative ileus**: physiological after laparotomy; recovery order = **small intestine within ≈ 24 h → stomach 24-48 h → COLON 48-72 h (last to recover)**. Diagnose return of function by **passage of flatus/stool** (bowel sounds are unreliable). **Paralytic ileus** beyond 3-5 days, or with vomiting and distension, is pathological — look for **hypokalaemia, hypomagnesaemia, opioids, anticholinergics, peritonitis, anastomotic leak, retroperitoneal haematoma**. Treat: NBM/NG decompression, correct electrolytes, minimise opioids, mobilise, chewing gum and early feeding help.
- **Distinguish ileus from mechanical obstruction**: ileus — **absent** bowel sounds, generalised distension, gas throughout including the colon; obstruction — **colicky pain, high-pitched/tinkling bowel sounds, no gas beyond the obstruction**, often POD > 7 (adhesions).
- **Nutrition (ERAS)**: **early oral feeding within 24 h** is safe after most surgery and reduces complications; enteral is always preferred to parenteral (‘**if the gut works, use it**’); TPN only when the gut cannot be used for > 7 days. Watch for **refeeding syndrome** (↓ PO₄, ↓ K⁺, ↓ Mg²⁺) in the malnourished.
- **Constipation**: almost universal with opioids — prescribe a **stimulant + softener laxative prophylactically**, encourage fluids, fibre and mobilisation.
- **Urinary retention**: commonest after **anorectal, hernia, pelvic and spinal-anaesthetic** surgery, in older men and with opioids/anticholinergics. Recognise by suprapubic fullness, restlessness, overflow dribbling, **bladder scan > 500-600 mL**; try privacy, standing/sitting position, running water, warm compress, analgesia → then **in-and-out catheterisation**.

C9. Post-Operative VTE, Mobilisation and Respiratory Care

- **DVT**: peak POD **5-10**; often **clinically silent**; calf pain/tenderness, unilateral swelling (> 3 cm difference), warmth, pitting oedema, dilated superficial veins. (Homans' sign is unreliable and should not be elicited.) Diagnose with **compression ultrasound + D-dimer/Wells score**.
- **PE**: sudden dyspnoea, pleuritic chest pain, tachycardia, tachypnoea, hypoxia, haemoptysis, syncope, unexplained hypotension. **Massive PE is a leading cause of preventable post-operative death**. Investigate with **CTPA**; treat with anticoagulation ± thrombolysis.
- **Prevention** = mechanical (early ambulation, ankle exercises, TED stockings, intermittent pneumatic compression) + pharmacological (LMWH), plus **adequate hydration and analgesia so the patient can move** — see §A7.
- **Respiratory care bundle**: semi-Fowler's position, **deep breathing and incentive spirometry 10 breaths hourly while awake, splinted (pillow-supported) coughing**, adequate analgesia (a patient who cannot cough because of pain will get atelectasis), early ambulation **within 24 h**, chest physiotherapy, oral hygiene (reduces ventilator-associated and post-op pneumonia), smoking cessation, humidified oxygen, nebulisers.

Complication	Timing & features	Management
Hypertrophic scar vs keloid	Hypertrophic — raised but stays within the original wound margins, tends to regress. Keloid — extends beyond the original margins, does not regress, recurs after excision; commoner in darker skin, over the sternum, shoulder and ear lobe	Silicone sheeting, pressure, intralesional steroid, excision + adjuvant therapy for keloid
Wound sinus / fistula	Persistent discharge; a retained suture, foreign body, necrotic tissue or a communication with a viscus (or a retained sponge — gossypiboma)	Investigate (imaging, sinogram), excise the tract, remove the cause

C7. Drains, Tubes and Catheters

Device	Type	Points
Penrose	Passive, open	Soft latex; drains by gravity/capillary action; higher infection risk
Corrugated / sheet drain	Passive, open	Retro-peritoneal and abscess cavities
Jackson-Pratt (JP)	Closed, active (bulb suction)	Bulb compressed to create low negative pressure; measure and record output; re-compress the bulb after emptying (a fully expanded bulb means the suction is lost)
Hemovac / Redivac	Closed, active (spring or high vacuum)	Used after orthopaedic, breast and neck surgery; Redivac is high-vacuum
Sump / double-lumen	Active	Air vent prevents tissue being sucked into the drain
Chest (intercostal) drain	Closed, underwater seal	Keep the bottle below the level of the chest at all times; never clamp a bubbling drain; ‘swinging’ with respiration = patent; continuous bubbling = air leak. If it disconnects → place the end in sterile water; if it falls out → cover with an occlusive dressing taped on 3 sides
Nasogastric tube	Decompression / feeding	Confirm position (pH < 5.5 on aspirate, or X-ray — the auscultation ‘whoosh’ test is unreliable and unsafe); record aspirate volume and character; routine NG use after laparotomy is no longer recommended (ERAS)
Urinary (Foley) catheter	Closed drainage	Aseptic insertion, closed system, bag always below bladder level and off the floor, no dependent loops, no routine bladder irrigation, remove within 24 h. CAUTI is the commonest healthcare-associated infection. Report output < 0.5 mL/kg/h (< 30 mL/h in an adult)

- **Purposes of a drain:** evacuate an existing collection, prevent accumulation of blood/serum/pus/bile/urine in a dead space, warn of a leak (‘sentinel’ drain), and allow a controlled tract to form.
- **Nursing responsibilities:** secure to prevent traction, keep patent and unkinked, measure and chart the **volume, colour and character** each shift, aseptic dressing at the exit site, observe for bleeding or enteric/bile content, and **never advance a drain back in**. Remove when output is minimal (typically < 25–50 mL/24 h) or as ordered.
- **Bright red brisk drain output → haemorrhage; faecal or bilious output → anastomotic leak; cloudy/foul → infection** — report immediately.

B13. Emergencies and Disasters — protocols, drugs and doses

Cardiopulmonary arrest in the OR

Item	Current (AHA/ILCOR) numbers
Sequence	C-A-B (Compressions – Airway – Breathing); high-quality CPR is the priority
Compression rate	100-120 per minute
Compression depth	≥ 5 cm (2 in) and ≤ 6 cm in adults; ⅓ of chest depth in children/infants (≈ 5 cm / 4 cm)
Compression : ventilation	30 : 2 (single/two rescuer adult, unintubated); continuous compressions with 1 breath every 6 s once an advanced airway is in place; 15 : 2 for two-rescuer paediatric
Chest-compression fraction & interruptions	Aim > 60-80 % ; interruptions < 10 s ; full recoil between compressions; change compressor every 2 min
Defibrillation	Biphasic 120-200 J (per manufacturer) or monophasic 360 J ; shockable rhythms = VF and pulseless VT; PEA and asystole are NOT shockable
Drugs	Adrenaline (epinephrine) 1 mg IV/IO every 3-5 min • Amiodarone 300 mg bolus then 150 mg (or lidocaine 1–1.5 mg/kg) for refractory VF/pVT • treat reversible causes
Reversible causes	4 H's – Hypoxia, Hypovolaemia, Hypo/Hyperkalaemia & metabolic, Hypothermia; 4 T's – Tension pneumothorax, Tamponade, Toxins, Thrombosis (coronary/pulmonary)
Monitoring quality	Waveform capnography — EtCO₂ < 10 mmHg indicates poor-quality CPR; a sudden rise suggests ROSC
Paediatric	PALS competency required for staff caring for children (your PDF); defibrillation 2 J/kg, then 4 J/kg; adrenaline 0.01 mg/kg
Team logistics in the OR	Emergency signal (flashing red light) operable from each OR • crash cart brought in by nursing staff • anaesthetist and surgeon lead • scribe documents every drug, dose, route, time and intervention — undocumented care is legally non-existent

Crash cart — minimum contents (asked as ‘which is NOT present’)

Defibrillator with paddles/pads + ECG monitor • Airway drawer: oral/nasal airways, laryngoscope with blades and spare bulbs/batteries, ET tubes of all sizes with stylet, LMAs, Magill forceps, suction catheters and a working suction unit, bag-valve-mask (Ambu) with reservoir, oxygen cylinder with flowmeter • Drugs (in date): adrenaline, atropine, amiodarone, lidocaine, adenosine, sodium bicarbonate, calcium chloride/gluconate, magnesium sulphate, dopamine/noradrenaline, dextrose 25/50 %, naloxone, flumazenil, hydrocortisone, chlorpheniramine, furosemide, GTN, midazolam • IV cannulae, syringes/needles, IV fluids and giving sets, intraosseous needle • CVP set, chest-drain set, cricothyrotomy set • backboard, gloves, sharps container, documentation sheet • checked and re-sealed after every use and at least daily/each shift, with a signed checklist.

Malignant hyperthermia (MH) — full protocol

Item	Detail
Genetics	Autosomal dominant ; mutation of the ryanodine receptor RYR1 gene on chromosome 19q ; uncontrolled Ca ²⁺ release from the sarcoplasmic reticulum → sustained muscle contracture and a hypermetabolic crisis
Triggers	All volatile inhalational agents (halothane, isoflurane, sevoflurane, desflurane, enflurane) and the depolarising relaxant SUCCINYLCHOLINE. Safe drugs: propofol, thiopentone, benzodiazepines, opioids, nitrous oxide, non-depolarising relaxants, all local anaesthetics

Item	Detail
Earliest / most sensitive sign	Unexplained rise in EtCO₂ (with tachypnoea/tachycardia); masseter spasm/trismus after succinylcholine is the classic early clinical clue. Hyperthermia is a LATE sign — do not wait for it
Full picture	Tachycardia, tachypnoea, rigidity , hypercarbia, hypoxaemia, mixed acidosis, hyperkalaemia , arrhythmias, rising temperature (may rise 1-2 °C every 5 min), myoglobinuria/rhabdomyolysis (cola-coloured urine) , raised CK, DIC, renal failure
Immediate managemene nt	1 Call for help and the MH cart; 2 STOP all triggering agents , change to a clean circuit/soda lime, hyperventilate with 100 % O₂ at ≥ 10 L/min ; 3 DANTROLENE 2.5 mg/kg IV bolus, repeated every 5-10 min until the crisis resolves — up to 10 mg/kg (may need 30 mg/kg) ; 4 abandon/finish surgery as fast as possible; 5 active cooling — iced IV saline, surface ice packs, gastric/bladder/wound lavage with cold saline; stop cooling at 38 °C to avoid overshoot; 6 treat hyperkalaemia (insulin–dextrose, calcium, bicarbonate), arrhythmias (avoid calcium-channel blockers with dantrolene — risk of hyperkalaemia and cardiac arrest), acidosis; 7 maintain urine output > 1-2 mL/kg/h (fluids, mannitol, furosemide) to protect the kidneys; 8 monitor core temperature, ABG, K ⁺ , CK, coagulation, urine myoglobin; 9 ICU observation ≥ 24-36 h (recrudescence in up to 25 %); 10 counsel the patient and family, refer for in-vitro contracture (caffeine-halothane) testing or genetic testing , and issue an alert card
Dantrolene practicalities	Supplied as 20 mg lyophilised vials reconstituted with 60 mL sterile water (newer formulation Ryandex 250 mg/vial); a 36-vial stock is recommended; acts by blocking the RYR1 channel. Documentation of all drugs, lines and treatments is mandatory (your PDF)
Mortality	> 70-80 % untreated → < 5-10 % with prompt dantrolene

Anaphylaxis in the OR

- Commonest triggers in the OR: **neuromuscular blocking agents, antibiotics (β-lactams), chlorhexidine, LATEX, colloids, contrast media**. **Latex anaphylaxis is typically delayed 20-60 min after exposure** (rather than immediate), because absorption is across mucosa/peritoneum — a favourite MCQ.
- Signs under anaesthesia: **unexplained hypotension and tachycardia (commonest), high airway pressure/bronchospasm, desaturation, rash/flushing, angio-oedema, cardiac arrest**.
- Management:** stop the trigger → call for help → **100 % O₂, secure the airway → ADRENALINE — 0.5 mg (0.5 mL of 1:1000) IM into the antero-lateral thigh, repeat every 5 min, or 50 µg IV boluses titrated (1:10 000, 0.5 mL)** if monitored and experienced, then infusion → **rapid IV crystalloid 20 mL/kg**, legs elevated → second line: **chlorpheniramine 10 mg IV, hydrocortisone 200 mg IV, salbutamol nebulisation**, glucagon if on β-blockers.
- Take **mast-cell tryptase** samples (immediately, 1-2 h and 24 h) and refer for allergy testing; document and give the patient a written record.
- Latex-safe environment:** first case of the day, latex-free gloves/tubing/catheters/drains, drug ampoules not vials with rubber bungs, dedicated **latex-free crash cart** (adrenaline, diphenhydramine, steroids, H₂ blockers, salbutamol, glucagon, latex-free airways and tubing), signage on the door. **Cross-reacting foods: banana, avocado, chestnut, kiwi** (also papaya, passion fruit, tomato, potato). **Highest-risk patients: spina bifida and children with multiple operations, plus health-care workers**.

- epithelialisation (contaminated/infected wounds, abscess cavities) → broad scar. **Tertiary / delayed primary closure:** left open 3-5 days then closed once contamination is controlled.
- Factors delaying healing** — local: **infection, ischaemia/poor perfusion, haematoma/seroma, foreign body, tension, dead space, radiation, movement, poor apposition**; systemic: **age, diabetes, malnutrition (protein, vitamin C → collagen cross-linking, vitamin A, zinc), anaemia and hypoxia, obesity, smoking, steroids and immunosuppressants, chemotherapy, jaundice, uraemia, malignancy**.
 - Dressing:** the first dressing over a primarily closed wound is usually left undisturbed for **24-48 h** (epithelialisation is complete by then and it forms a bacterial barrier); thereafter change using **aseptic (sterile) technique**, inspect for erythema/discharge, and document. Change immediately if soaked (strike-through).
 - Suture removal times** — face/neck **3-5 days**; scalp **7-10**; trunk/abdomen **7-10 (up to 12)**; upper limb **10-14**; lower limb **10-14**; over joints, back, palms and soles **14 days**; children heal faster (shorter times), the elderly/steroid users need longer.

Wound complications

Complicati on	Timing & features	Management
Haematoma	Within 24–48 h; swelling, discoloration, pain, may compress (neck haematoma after thyroidectomy = airway emergency)	Small → observe; expanding/compressive → evacuate, open the wound (bedside for neck), secure haemostasis
Seroma	Days 3–10, especially after mastectomy, axillary/groin dissection, hernia mesh ; fluctuant, painless swelling	Aspirate under aseptic technique, compression, drains; may recur
Surgical site infection	POD 5–7 ; pain, erythema, warmth, induration, purulent discharge, fever, wound edge separation. (Very early, POD 1–2, spreading cellulitis/crepitus → suspect β-haemolytic streptococcus or clostridial/necrotising infection — surgical emergency)	Open, drain and culture; debride; pack for healing by secondary intention; antibiotics guided by culture — ‘the treatment of pus is drainage’
Wound dehiscence	POD 5–10 (classically day 7–8) ; the herald sign is a sudden discharge of serosanguineous (‘salmon-pink’) fluid from the wound and a palpable gap; patient may report a ‘giving-way’ sensation on coughing	Reinforce with an abdominal binder, avoid straining, inform the surgeon → usually re-suturing in theatre with tension sutures
Evisceratio n (burst abdomen)	Protrusion of viscera through the wound; often follows a cough or straining	EMERGENCY: stay with the patient, place them supine with knees flexed (low Fowler’s), COVER the viscera with sterile gauze soaked in warm sterile normal saline (never dry gauze, never attempt to push the bowel back), keep NBM, IV access and fluids, monitor for shock, analgesia, notify the surgeon and prepare for immediate theatre.
Incisional hernia	Weeks–years later, a late consequence of a subclinical dehiscence or infection	Elective repair, usually with mesh

Onset	“W”	Cause	Clues & action
POD 3-5	Water	Urinary tract infection (catheter-associated)	Dysuria/suprapubic pain, urinalysis + culture; remove or change the catheter , antibiotics
POD 4-6	Walking (‘Wein’)	Deep-vein thrombosis / pulmonary embolism	Calf pain/swelling, unilateral oedema, sudden dyspnoea, pleuritic pain, tachycardia, hypoxia. Doppler/CTPA; anticoagulate
POD 5-7	Wound	Surgical site infection; also anastomotic leak (POD 5-7, with tachycardia and ileus)	Inspect — erythema, tenderness, induration, discharge; open and drain, culture, antibiotics ; suspect a leak if the patient is 'not doing well' with tachycardia
POD > 7	Wonder drugs & Wou nder-things	Drug fever (antibiotics, heparin), abscess/collection, C. difficile colitis, thrombophlebitis at the IV site, central-line infection, sinusitis, acalculous cholecystitis, transfusion-related	Review the drug chart, culture blood/urine/line tips, CT for a collection

MNEMONIC — “W**ind, **W**ater, **W**ound, **W**alking, **W**onder drugs (+ **W**hat we did / **W**onder about lines)”**

Sequence in time: **Wind (1-2) → Water (3-5) → Walking/Wein (4-6) → Wound (5-7) → Wonder drugs (> 7).**

A fever in the first 24 h is usually NOT infective — it is the inflammatory/cytokine response to surgery, atelectasis, a drug or transfusion reaction, or malignant hyperthermia.

Fever after POD 5 with wound signs = infection until proved otherwise.

C6. Wound Healing, Wound Care and Wound Complications

Phases of wound healing

Phase	Timing	Events	Dominant cell
Haemostasis	Immediate (minutes-hour s)	Vasoconstriction, platelet plug, fibrin clot	Platelets
Inflammatory (lag/substrate) phase	Day 0-4/5	Vasodilatation, exudate, phagocytosis, debridement. Clinically rubor, calor, tumor, dolor ; wound held only by the clot and sutures	Neutrophils (24-48 h) → Macrophages (48-72 h — the key orchestrating cell)
Proliferative (fibroplasia / granulation) phase	Day 4-21	Fibroblast migration, collagen III deposition , angiogenesis, granulation tissue, epithelialisation, wound contraction	Fibroblasts (+ endothelial cells, myofibroblasts)
Maturation / remodelling phase	Day 21 → up to 1-2 years	Collagen III replaced by collagen I , cross-linking, scar becomes pale and flat, tensile strength rises	Fibroblasts, collagenase (MMPs)

- **Tensile strength:** ≈ 5-10 % at 1 week, 30-50 % at 1 month, and only ever reaches 70-80 % of unwounded skin. This is why the wound is at greatest risk of dehiscence in the **first week** and why heavy lifting is restricted for 6 weeks.
- **Types of healing — Primary (first) intention:** clean wound, apposed edges, minimal granulation, fine scar. **Secondary intention:** wound left open, heals by granulation, contraction and

Local anaesthetic systemic toxicity (LAST) — a modern must-know

Progression: perioral tingling, metallic taste, tinnitus, visual disturbance, slurred speech, agitation → **seizures** → CNS depression → **arrhythmia and cardiac arrest** (bupivacaine is the most cardiotoxic). **Management: stop injecting, 100 % O₂, control seizures with a benzodiazepine (avoid propofol in cardiovascular instability), and give LIPID EMULSION 20 % — 1.5 mL/kg bolus over 1 min then 0.25 mL/kg/min infusion;** prolonged CPR may be required; **avoid vasopressin, calcium-channel blockers, β-blockers and further local anaesthetic.**

Fire in the operating room

Element	Detail
Fire triangle	FUEL + OXIDISER + IGNITION (heat) source — all three are abundant in an OR (some texts add a 4th: the chemical chain reaction → ‘fire tetrahedron’)
Commonest ignition source	Electrosurgical unit (≈ 70-90 % of OR fires) , then the laser, then fibre-optic light cables and defibrillators
Commonest fuel	Alcohol-based skin preparation solutions (pooled or not dry), then drapes, gowns, gauze, hair, plastic/rubber, ET tubes, GI bowel gas
Oxidiser	Oxygen and nitrous oxide — the OR is an ‘oxygen-enriched atmosphere’; O ₂ pools under drapes near the head (the classic head-and-neck surgery fire)
Prevention	Let alcohol prep dry completely (≥ 3 min) , never allow pooling, blot puddles, remove wet linen; keep FiO ₂ < 30 % for head/neck cases and use a sealed airway; pause oxygen before activating the ESU/laser in the airway ; moisten sponges, drapes and hair near the site; keep the pencil holstered; use laser-safe ET tubes with saline-filled cuffs; scavenge oxygen from under the drapes
Response — R A C E	Rescue/Remove the patient from danger • Alarm (activate the alarm and call the code, e.g. CODE RED) • Confine/Contain (close doors, shut off medical gases) • Extinguish or Evacuate
Extinguisher use — P A S S	Pull the pin • Aim at the base of the fire • Squeeze the handle • Sweep side to side
Extinguisher classes	A = ordinary combustibles (paper, cloth) - water/foam • B = flammable liquids - CO ₂ /dry chemical • C = electrical - CO ₂ or dry chemical (never water) • D = combustible metals • K = kitchen fats. The OR normally stocks a CO₂ or ABC dry-chemical extinguisher; CO₂ is preferred for electrical/equipment fires because it leaves no residue. (Halon 1301 was the historical OR agent)
Airway fire drill	1 Stop ventilation and disconnect the oxygen/circuit immediately ; 2 remove the endotracheal tube and any burning material; 3 pour saline into the airway; 4 re-establish ventilation on air, 5 re-intubate and bronchoscope to assess thermal injury, 6 steroids/ICU care. Then preserve the equipment, file an incident report, and report the device to the regulator
Drills	Fire drills at least twice a year, unannounced ; every employee must know the code name, the location of the alarm, extinguishers, medical-gas shut-off valves and the evacuation route. Drapes and gowns are fire-RETARDANT, not fire-PROOF — your PDF’s repeated caution

Mass-casualty / disaster preparedness — absent from your PDF

Concept	Content
Types	Internal (fire, power failure, gas failure, structural, IT/cyber, violent intruder) vs External (earthquake, flood, road/rail crash, bomb blast, chemical/radiation leak, epidemic)

Concept	Content
Framework	Hospital Emergency Incident Command System (HEICS/HICS) with a written Hospital Disaster Management Plan based on a Hazard Vulnerability Analysis (HVA) ; defined roles, call-tree, surge capacity, alternate site, mock drills at least annually
Triage (Fr. <i>trier</i> , ‘to sort’)	START = Simple Triage And Rapid Treatment (≤ 60 s per casualty), assessing RPM — Respiration, Perfusion, Mental status : walking wounded → GREEN ; RR > 30, absent radial pulse/capillary refill > 2 s, or not obeying commands → RED ; otherwise → YELLOW ; not breathing after airway opening → BLACK
Triage colour codes	RED = Priority I - immediate (life-threatening but salvageable: airway obstruction, tension pneumothorax, major haemorrhage, shock) • YELLOW = Priority II - delayed (can wait 1–4 h: long-bone fracture, abdominal injury without shock) • GREEN = Priority III - minor/ambulatory • BLACK = Priority 0 - dead / expectant (injuries incompatible with survival given available resources)
Key principle	In a mass casualty the aim shifts from ‘the best for each individual’ to ‘the greatest good for the greatest number’ — hence expectant categorisation, and re-triage at every stage (triage is dynamic)
Pharmacist's / medical assistant's disaster role	Maintain and rotate disaster drug stock and antidote kits (atropine–pralidoxime for organophosphate, potassium iodide for radiation, hydroxocobalamin/nitrites for cyanide, naloxone, antivenom, tetanus toxoid/Ig), prepare IV fluids and analgesics in bulk, control narcotics documentation, advise on drug–drug interactions and dosing in renal failure/crush injury, oversee cold chain, and manage donated-medicine screening
Power failure	Automatic emergency generator back-up (must start within 10 s and carry the OR, PACU, ICU and lifts); UPS for monitors, ventilators and the anaesthesia machine; every team member must know where the high-intensity torches are kept ; know the location of manual (self-inflating) ventilation bags and the reserve oxygen cylinder for medical-gas failure
Code colours (institution-specific — typical Indian hospital set)	Blue = cardiac arrest • Red = fire • Pink = infant/child abduction • Orange = mass casualty / external emergency • Yellow = internal emergency / bomb threat (varies) • Black = bomb threat (varies) • Brown = evacuation • Grey = violent person / security • Purple = hostage. Always answer per your own institution's published list.

PART C — POST-OPERATIVE PATIENT CARE (THE PDF'S BIGGEST OMISSION)

WHY THIS PART MATTERS

Your syllabus heading is “Preparation of Patient for Operation, **Pre & Post Operative Patient Care**”. Chapter 1 of the PDF ends at the sentence “the patient's care is transferred to the perioperative RN in the PACU, who now assumes the role of the patient's advocate” — and the post-operative record is only reproduced as a blank form. **Everything in Part C is therefore new material, and it is where the majority of clinical MCQs from this unit are set.**

C1. Phases of Post-Operative Care & the PACU

Phase	Location	Focus
Phase I — immediate recovery	PACU / post-anaesthesia care unit (recovery room), 1 nurse : 1-2 patients	Emergence from anaesthesia; airway, breathing, circulation, consciousness, temperature, pain, haemorrhage; continuous monitoring
Phase II — intermediate	Step-down/ambulatory unit or ward	Preparation for discharge home or to the ward; oral intake, ambulation, teaching

Element	Detail
Opioid adverse effects (*N-CROP*)	Nausea/vomiting • Constipation (does not develop tolerance → always prescribe a laxative) • Respiratory depression (sedation precedes it — monitor the sedation score, not just RR) • Obstruction/ileus and urinary retention • Pruritus , plus miosis, hypotension, tolerance
NSAID cautions	Renal impairment, hypovolaemia, GI bleeding/ulcer, asthma, elderly, bleeding risk, avoid after bariatric anastomoses and in some bone-graft/fusion surgery ; ketorolac max 5 days
Paracetamol	Maximum 4 g/24 h (3 g in the frail, low weight, liver disease or alcohol excess) — watch for duplication in combination products, a classic pharmacist intervention

C4. Post-Operative Nausea & Vomiting (PONV)

Item	Detail
Incidence	20-30 % overall; up to 70-80 % in high-risk patients. Commonest reason for unanticipated admission after day surgery, and the outcome patients fear most after pain
Apfel simplified risk score (adults)	1 point each: female sex • non-smoker • previous PONV or motion sickness • expected post-operative opioid use. Risk with 0/1/2/3/4 factors ≈ 10 / 20 / 40 / 60 / 80 %
Other risk factors	Younger age, general (vs regional) anaesthesia, volatile agents and nitrous oxide , duration of anaesthesia, neostigmine in high dose, type of surgery (gynaecological, laparoscopic, ENT/middle-ear, squint, breast, bariatric), gastric distension, hypotension, dehydration, pain, early mobilisation, opioids
Drug classes & receptors	5-HT₃ antagonists — ondansetron 4-8 mg IV (give at the end of surgery; SE: headache, constipation, QT prolongation), palonosetron • Corticosteroid — dexamethasone 4-8 mg IV (give at induction; also improves analgesia; watch glucose) • D₂ antagonists — metoclopramide (prokinetic; extrapyramidal effects, avoid in obstruction), droperidol (QT), domperidone • NK₁ antagonist — aprepitant • Antihistamine — cyclizine, promethazine • Anticholinergic — hyoscine (scopolamine) patch • Others — propofol TIVA (antiemetic), acupressure at P6 (Neiguan)
Strategy	Risk-adapted: 1-2 risk factors → 2 prophylactic agents from DIFFERENT classes; ≥ 3 → 3-4 agents + TIVA with propofol, avoid nitrous oxide and volatile agents, minimise opioids (regional/multimodal), ensure adequate hydration. For breakthrough PONV, use a drug from a class not already given
Nursing measures	Nil by mouth until nausea settles then sips, avoid sudden movement, oral hygiene, decompress the stomach with an NG tube if distended , treat pain and hypotension, correct electrolytes, IV fluids, cool cloth, position to avoid aspiration

C5. Post-Operative Fever — the “6 W's” timeline (very high yield)

Onset	“W”	Cause	Clues & action
Immediate — during or within hours	Wonder drugs / What did we do?	Drug or transfusion reaction, malignant hyperthermia , pre-existing infection, trauma of surgery (cytokine response), thyroid storm	MH → rigidity + ↑ EtCO ₂ → dantrolene. Stop the transfusion for any reaction
POD 0-2 (24-48 h)	Wind	ATELECTASIS (commonest cause of early post-op fever) ; aspiration, early pneumonia	Low-grade fever, ↓ basal breath sounds. Incentive spirometry, deep breathing, ambulation, analgesia, physiotherapy — not antibiotics

Problem	Key facts
Post-operative cognitive dysfunction (POCD) / post-op delirium	Elderly, pre-existing cognitive impairment, major/cardiac surgery, sepsis, sleep disruption, anticholinergic burden. Prevent: orientation, glasses/hearing aids, sleep hygiene, avoid benzodiazepines and anticholinergics, treat pain and constipation, early mobilisation
Hypothermia (core < 36 °C)	Causes: cold OR, anaesthetic-induced vasodilatation and loss of thermoregulation, cold fluids/irrigation, large exposed cavity, elderly/neonates, prolonged surgery. Consequences: shivering (↑ O₂ consumption up to 400 %), coagulopathy and ↑ blood loss, ↑ SSI, arrhythmia and cardiac events, delayed drug metabolism and delayed emergence, patient discomfort. Prevent/treat: forced-air warming, warmed IV fluids and irrigation, warmed blankets, raised room temperature, heat-moisture exchanger, limit exposure
Shivering	Very common; treat by warming plus pethidine (meperidine) 12.5-25 mg IV (most effective), clonidine, dexmedetomidine, tramadol, ondansetron; give oxygen
Malignant hyperthermia	Can present in the PACU as well as intra-operatively — see §B13
Nerve injury / paralysis	Examine and document limb movement, power and sensation on arrival; paraesthesia or inability to move a limb → exclude positioning injury, epidural haematoma (back pain + bilateral weakness + urinary retention = EMERGENCY, needs urgent MRI and decompression within 8 h) and compartment syndrome

C3. Post-Operative Pain Management

Element	Detail
Assessment	Self-report is the gold standard: Numeric Rating Scale 0-10, Visual Analogue Scale, Wong-Baker FACES scale (children ≥ 3 y), FLACC (Face, Legs, Activity, Cry, Consolability — pre-verbal children), FPS-R ; behavioural/physiological cues in the unconscious. Assess at rest and on movement, and re-assess after each intervention
Consequences of untreated pain	Splinting → atelectasis and pneumonia ; immobility → DVT ; sympathetic surge → tachycardia, hypertension, ischaemia; ileus; urinary retention; delirium; poor sleep; chronic post-surgical pain
Multimodal analgesia (the modern standard)	Combining drugs with different mechanisms lowers each dose and its side-effects: paracetamol 1 g q6h + an NSAID (diclofenac/ketorolac) + local/regional block ± opioid ± adjuvants (gabapentinoid, ketamine, dexmedetomidine, dexamethasone, IV lignocaine, magnesium)
WHO analgesic ladder	Step 1 non-opioid ± adjuvant → Step 2 weak opioid (tramadol, codeine) ± non-opioid → Step 3 strong opioid (morphine, fentanyl). For acute post-operative pain the ladder is used downwards (start strong, step down)
Routes	IV titration in PACU → PCA (patient-controlled analgesia) → oral once tolerating. IM injections are discouraged (painful, erratic absorption). Avoid the oral route until bowel function returns after major GI surgery
PCA	Typical morphine PCA: bolus 1 mg, lockout 5-10 min, no background infusion in opioid-naïve adults; the patient must be awake to press the button — the built-in safety feature; never allow a relative to press it ; monitor sedation score and RR
Regional techniques	Epidural (thoracic for laparotomy/thoracotomy — best dynamic pain relief, reduces pulmonary complications and ileus; watch for hypotension, motor block, urinary retention, epidural haematoma/abscess), spinal opioids, TAP block, rectus-sheath, paravertebral, brachial plexus, femoral/adductor canal, erector-spinae blocks, wound infiltration and catheters

Phase	Location	Focus
Phase III — extended / convalescent	Ward, day-care lounge, home	Rehabilitation, wound care, complication surveillance, follow-up

PACU hand-over — what must be communicated (SBAR / ‘ISBAR’)

Patient's name, age and language • procedure performed and findings • surgeon and anaesthetist • type of anaesthesia and reversal given • airway (still intubated? LMA removed?) • allergies • relevant co-morbidity and medications • intra-operative course including vital-sign trends and any events • estimated blood loss, fluids and blood products given, urine output • drains, catheters, packs, dressings • analgesia and antiemetics given (drug, dose, time) • laterality and position of the limb/dressing • post-operative orders, expected concerns, and whom to call. The anaesthetist does not leave until the PACU nurse accepts the patient and the hand-over is documented.

PACU monitoring — what and how often

- Vital signs every 5-15 min** initially, then every 15 min until stable, then every 30 min: **RR/pattern, SpO₂, BP, pulse, temperature, ECG, EtCO₂ if intubated, pain score, level of consciousness, plus urine output and drain output hourly.**
- Position: **lateral or head-up 15-30°** if not contraindicated — keeps the tongue forward and allows secretions to drain (‘recovery position’); **supine flat** after spinal anaesthesia or hypotension.
- Oxygen** by mask/nasal prongs until awake and SpO₂ is adequate on air; **side rails up, never leave the patient unattended**, keep suction and an Ambu bag at the bedside.
- Check: **dressing (ooze/soakage), surgical site, drain patency and character, IV site, peripheral pulses and capillary refill distal to any cast/tourniquet, ability to move limbs and any paraesthesia (rule out positioning injury and assess spinal regression), bladder distension.**
- Warming** — forced-air warming blanket, warmed IV fluids, raised room temperature; aim core temperature **> 36 °C** before discharge from PACU.

Aldrete Score (Post-Anaesthesia Recovery Score) — memorise this table

Parameter	2 points	1 point	0 points
Activity (voluntary movement)	Moves all four limbs on command	Moves two extremities	Unable to move
Respiration	Breathes deeply and coughs freely	Dyspnoeic, shallow or limited breathing	Apnoeic / needs ventilation
Circulation (BP)	BP ± 20 % of pre-anaesthetic level	BP ± 20-49 %	BP ± ≥ 50 %
Consciousness	Fully awake	Rousable on calling	Not responding
Oxygen saturation	SpO₂ > 92 % on room air	Needs O ₂ to keep SpO ₂ > 90 %	SpO ₂ < 90 % even with O ₂

ALDRETE — EXAM POINTS
Maximum score = 10. Discharge from PACU requires ≥ 9 (with no single parameter scoring 0), **and** stable vital signs, controlled pain and PONV, no active bleeding, and normothermia.
Mnemonic: A-R-C-C-O (Activity, Respiration, Circulation, Consciousness, Oxygenation). The **original 1970 Aldrete score used skin COLOUR** as the fifth criterion; the **Modified Aldrete score replaced colour with pulse-oximetry SpO₂**.
Pain and nausea are NOT part of the Aldrete score — they belong to the **PADSS**. This inversion is a classic MCQ.
PADSS (Post-Anaesthesia Discharge Scoring System, for day-care/ambulatory discharge HOME) scores 0–2 each for: **vital signs • ambulation • nausea/vomiting • pain • surgical bleeding; ≥ 9 required for discharge home.** Modern ‘fast-tracking’ also requires: an escort, written instructions, ability to tolerate oral fluids, and — in most protocols — the **ability to void is no longer mandatory** except after spinal/anorectal/hernia surgery.
After spinal/epidural anaesthesia additionally document **return of motor power, sensation and proprioception**, and the **sensory level**.

C2. Immediate Post-Operative Complications — recognise and act

A. Respiratory

Complication	Cause	Recognition	Management
Airway obstruction (commonest immediate PACU emergency)	Tongue falling back against the posterior pharynx in the sedated supine patient; secretions, blood, laryngeal oedema, foreign body, dentures	Snoring/stridor, see-saw (paradoxical) chest movement, tracheal tug, no air movement, desaturation	Head tilt-chin lift / jaw thrust, lateral position, suction, oral/nasal airway, 100 % O ₂ ; re-intubate if needed
Laryngospasm	Stimulation of the cords during light anaesthesia, secretions, blood	Inspiratory stridor progressing to complete silence, paradoxical effort, desaturation	Remove the stimulus, suction, 100 % O₂ with CPAP and jaw thrust; if it persists — propofol, then succinylcholine 0.1-1 mg/kg and intubate
Hypoventilation / respiratory depression	Residual anaesthetic, opioids, incomplete reversal of neuromuscular blockade, splinting from pain, obesity, COPD, hypothermia	Slow shallow breathing, ↑ EtCO ₂ , ↓ SpO ₂ , drowsiness, pin-point pupils (opioid)	Stimulate, O ₂ , support ventilation; naloxone 40-80 µg IV increments for opioids; neostigmine/sugammadex for residual relaxant; flumazenil for benzodiazepines
Hypoxaemia	Atelectasis, hypoventilation, V/Q mismatch, diffusion hypoxia, aspiration, pulmonary oedema, pneumothorax	SpO ₂ < 90 %, restlessness (an early sign), tachycardia, cyanosis (late)	O ₂ , sit up, deep breathing/incentive spirometry, treat cause, chest X-ray
Aspiration	Full stomach, obtunded reflexes	Cough, wheeze, tachypnoea, desaturation, right-lower-lobe infiltrate	Head-down lateral, suction, O ₂ , bronchoscopy if particulate; steroids and prophylactic antibiotics are NOT routinely indicated

Complication	Cause	Recognition	Management
Atelectasis	Commonest post-operative pulmonary complication; shallow breathing, retained secretions, upper-abdominal/thoracic incision	Low-grade fever POD 1-2, decreased breath sounds, dull percussion at the bases, tachypnoea	Deep breathing + incentive spirometry (10 breaths/hour), early mobilisation, adequate analgesia, chest physiotherapy, humidified O₂

B. Cardiovascular

Problem	Common causes	Action
Hypotension	Hypovolaemia (commonest — blood loss, third-space loss, inadequate replacement), residual anaesthetic/vasodilatation, sympathetic block from spinal/epidural, opioids, arrhythmia, myocardial ischaemia, tension pneumothorax, tamponade, sepsis, anaphylaxis	Assess the surgical site and drains for bleeding first; head-down/legs up, O ₂ , rapid crystalloid bolus 250-500 mL (10-20 mL/kg) , vasopressor if needed, ECG, Hb, call surgeon/anaesthetist; re-exploration for surgical bleeding
Hypertension	Pain (commonest), bladder distension, hypoxia/hypercarbia, hypothermia with shivering, pre-existing hypertension with omitted drugs, fluid overload, emergence excitement	Treat the cause first — analgesia, catheterise/scan the bladder, oxygenate, warm — before reaching for antihypertensives (labetalol, GTN, hydralazine)
Arrhythmia	Sinus tachycardia (pain, hypovolaemia, fever, anaemia, anxiety, anticholinergics) • bradycardia (opioids, neostigmine, vagal traction, hypoxia, β-blockade, spinal to T1-T4) • AF/VT (electrolyte imbalance, ischaemia, hypoxia, catecholamines)	Correct hypoxia, hypovolaemia, K⁺/Mg²⁺, acidosis and pain; ECG; atropine for symptomatic bradycardia; treat per ACLS if unstable
Myocardial ischaemia/infarction	Peak incidence POD 0-3; often silent or atypical in the post-op period because of analgesia	Any unexplained hypotension, arrhythmia, dyspnoea or confusion after surgery → ECG + troponin; O ₂ , analgesia, cardiology referral
Haemorrhage / shock	Reactionary haemorrhage (within 24 h — slipped ligature, dislodged clot as BP rises and vasoconstriction wears off) • Secondary haemorrhage (POD 7-14 — infection eroding a vessel) • primary = during surgery	Class I-IV haemorrhagic shock: tachycardia and narrowed pulse pressure precede hypotension; tachycardia with a normal BP in a young patient is compensated shock — do not be reassured. Two large-bore cannulae, crystalloid, cross-matched blood / massive transfusion protocol (1:1:1 RBC:FFP:platelets), tranexamic acid, keep warm, surgical control of bleeding is definitive

C. Neurological & thermoregulatory

Problem	Key facts
Delayed emergence	Residual anaesthetic/opioid/relaxant • hypoglycaemia, hypoxia, hypercarbia, hyponatraemia, hypothermia (< 33 °C prevents awakening) • raised ICP, stroke, seizure • central anticholinergic syndrome. Check glucose, ABG, electrolytes and pupils; reverse opioids/benzodiazepines; think stroke if focal signs
Emergence delirium / agitation	Commonest in children and the elderly; also hypoxia (always exclude hypoxia first), pain, full bladder, ketamine, anticholinergics, alcohol/benzodiazepine withdrawal. Reassure, orientate, involve parents, treat pain; avoid physical restraint